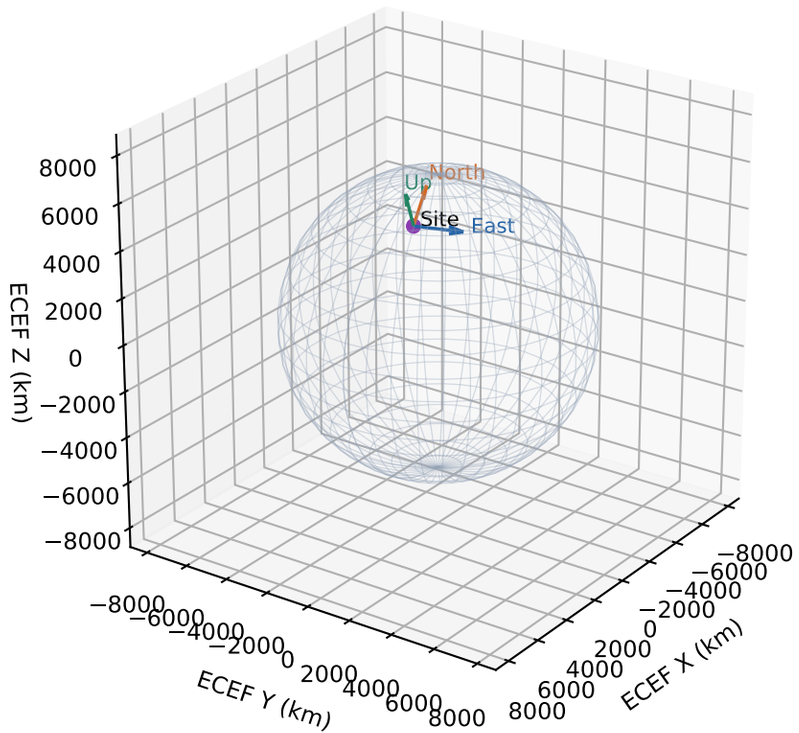
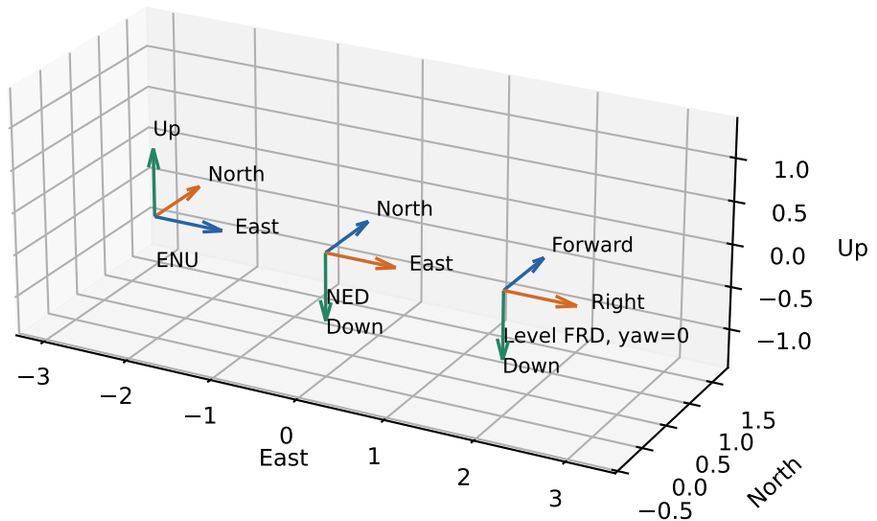
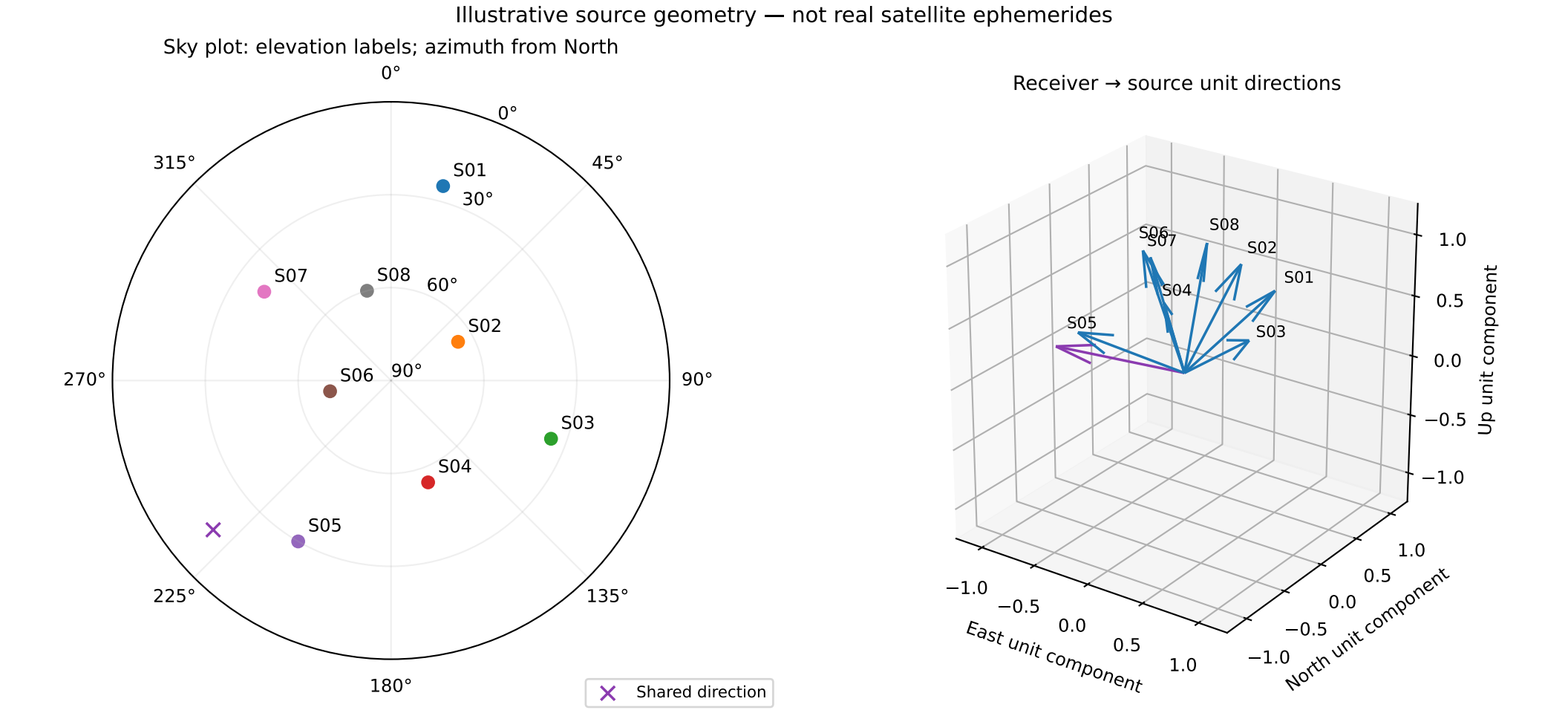
Coordinate conventions: latitude is geodetic; heights are ellipsoidal

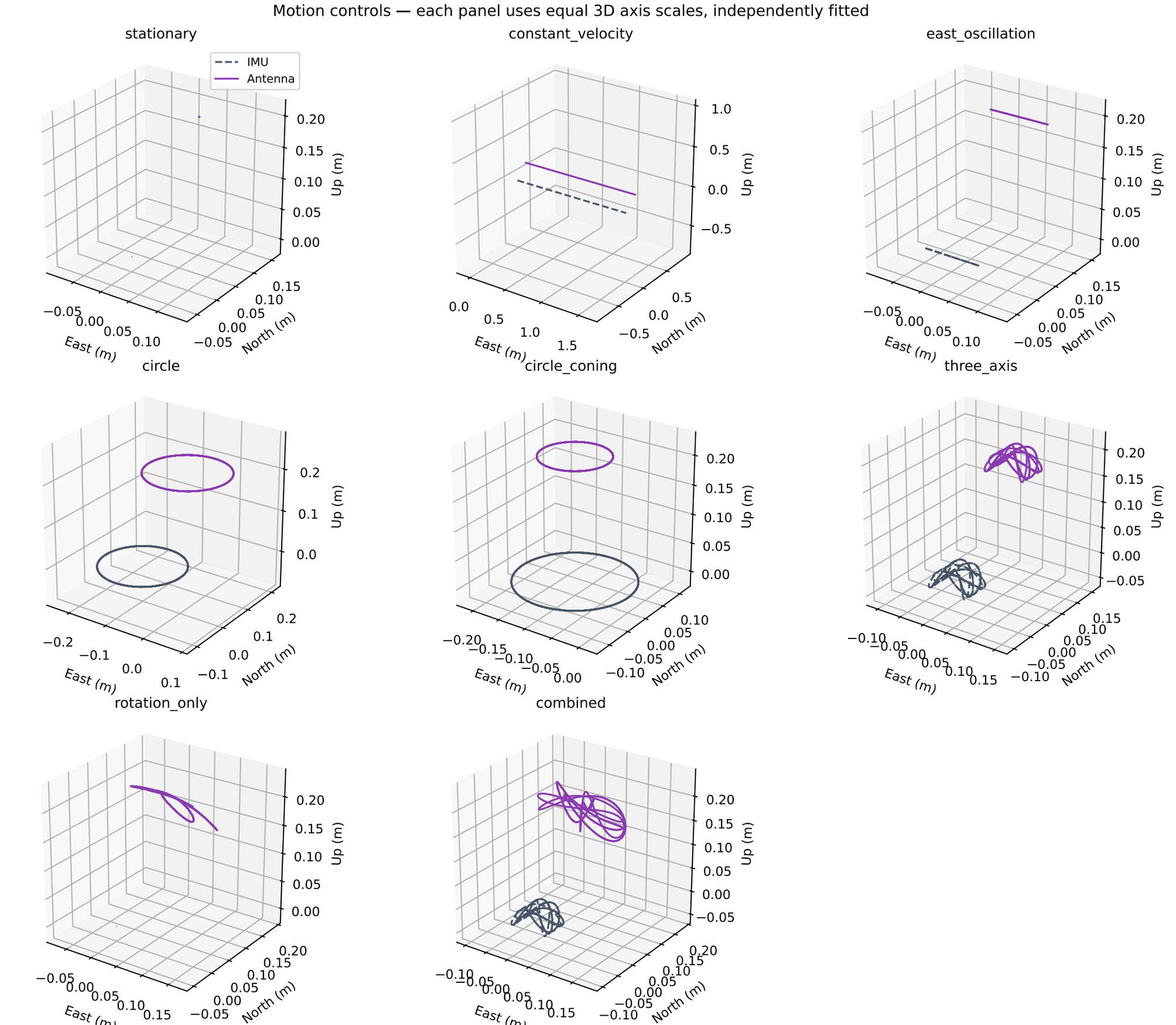
WGS-84 and the site tangent basis
Direction arrows enlarged for visibility

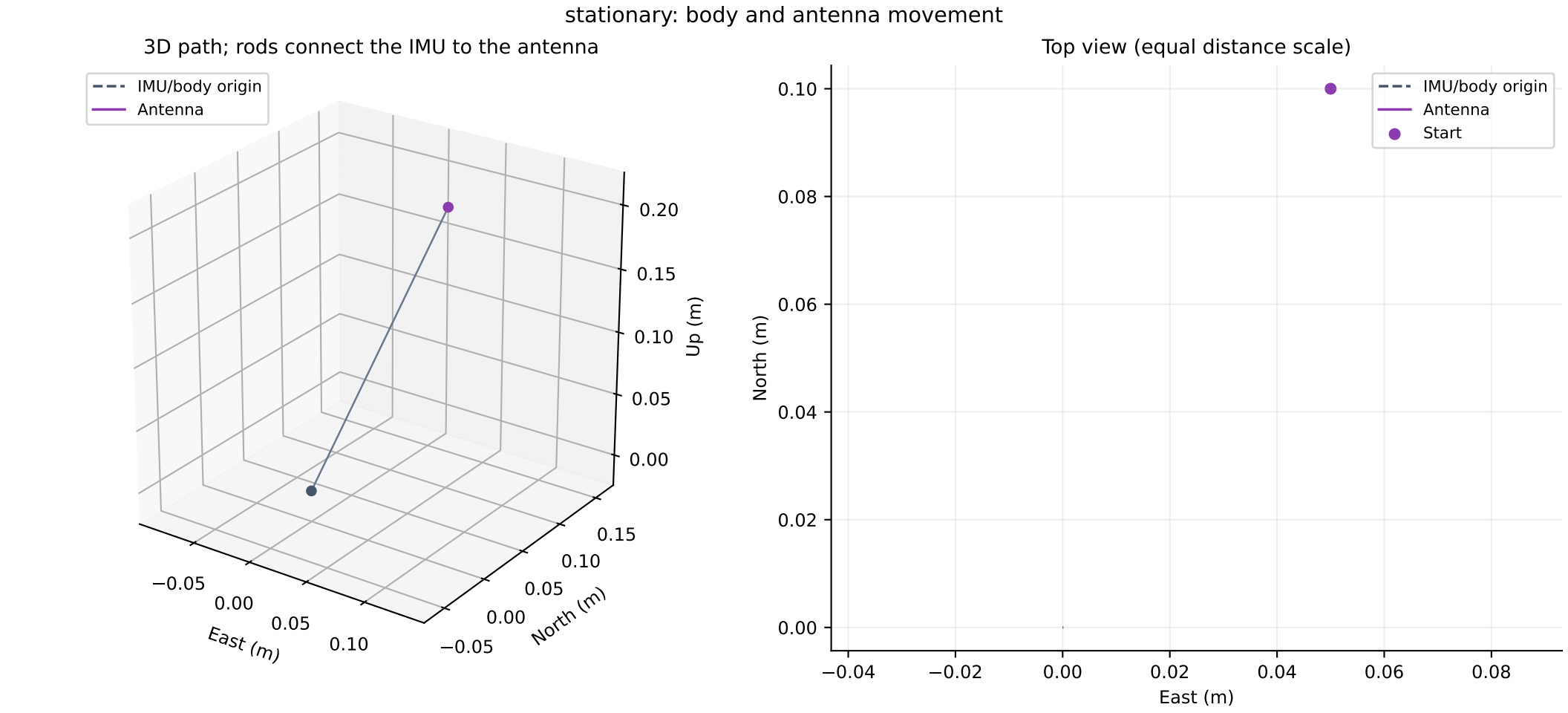


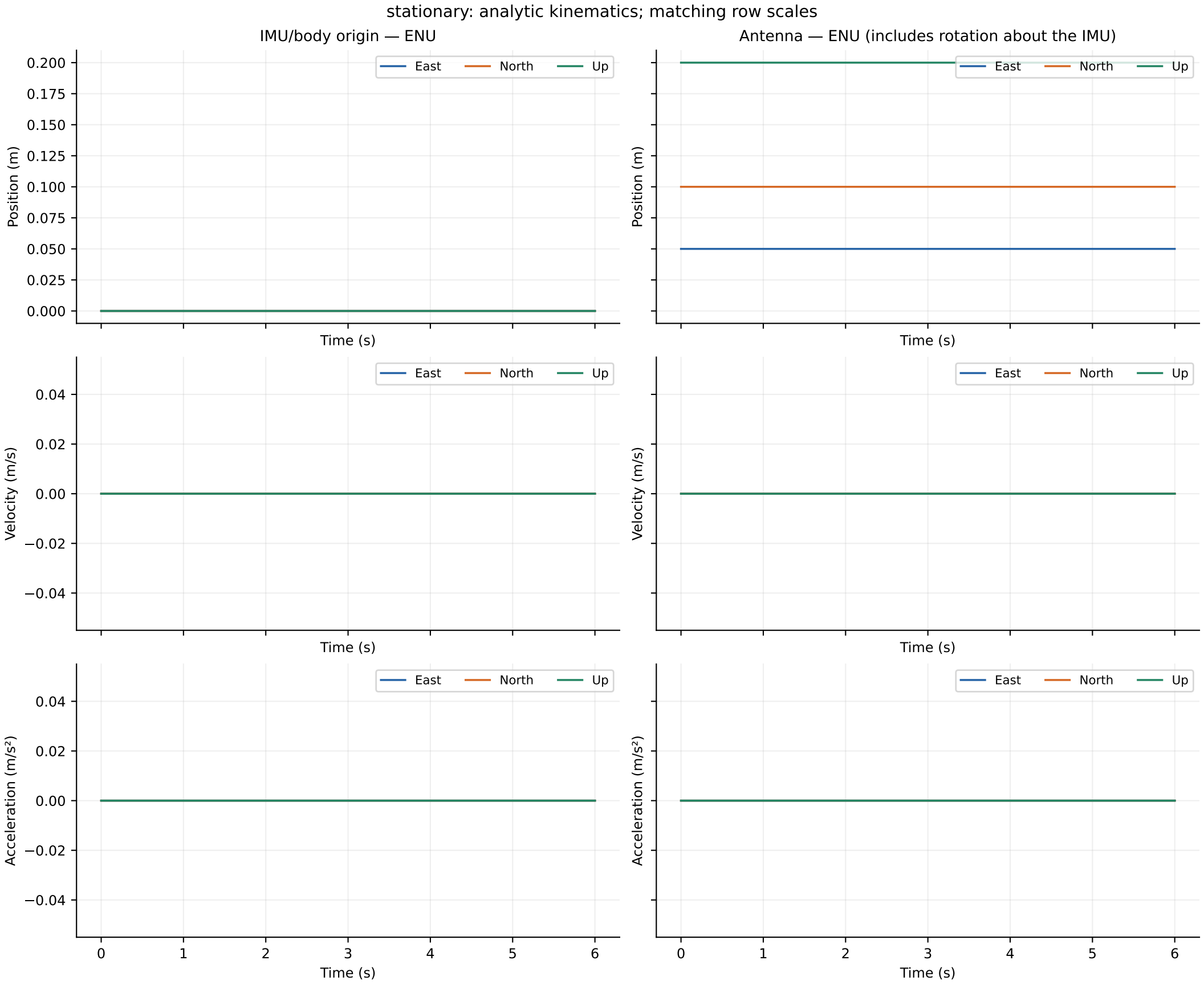
ENU, NED and body axes at zero attitude
Display origins separated; bases have unit scale

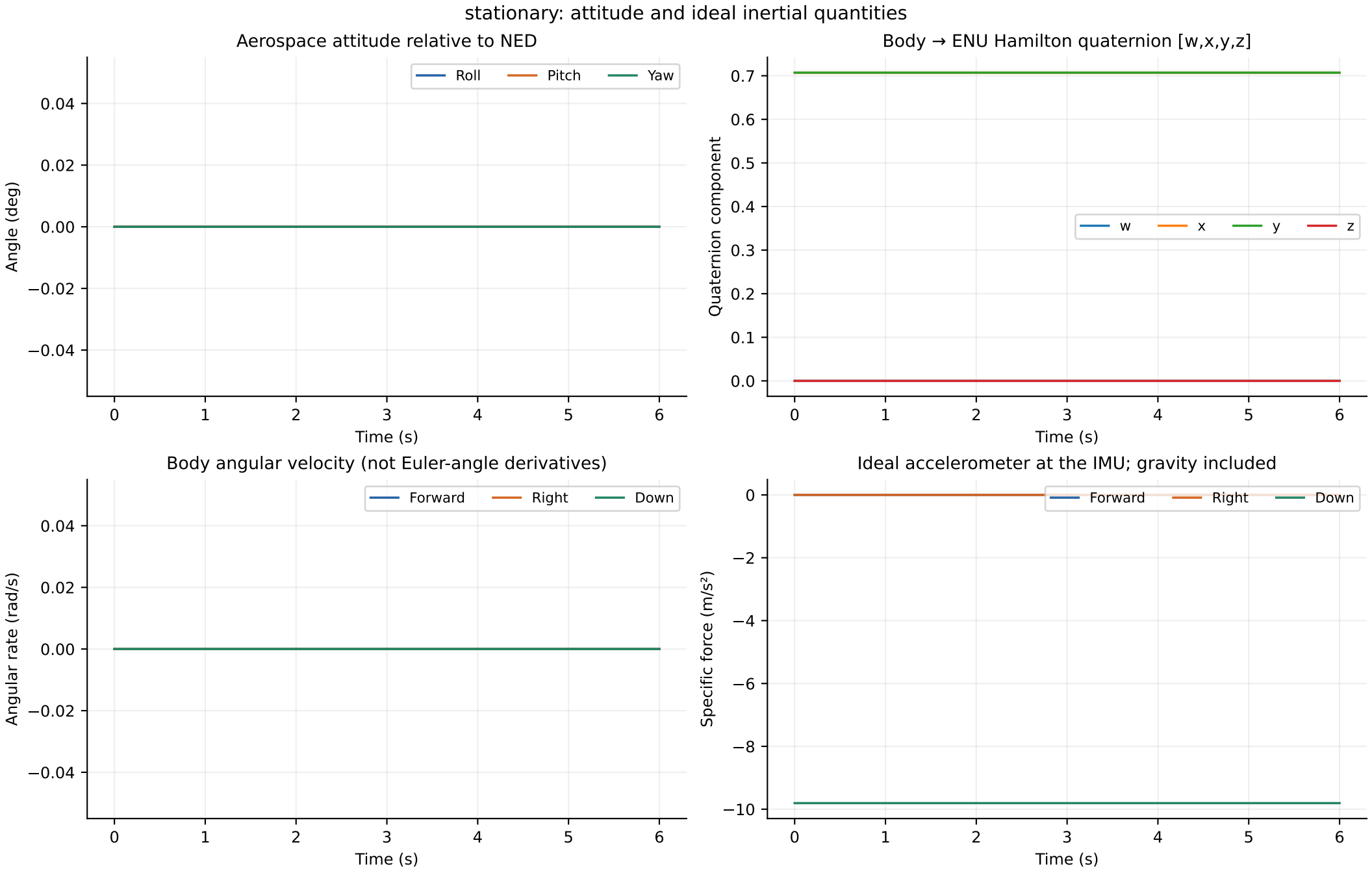


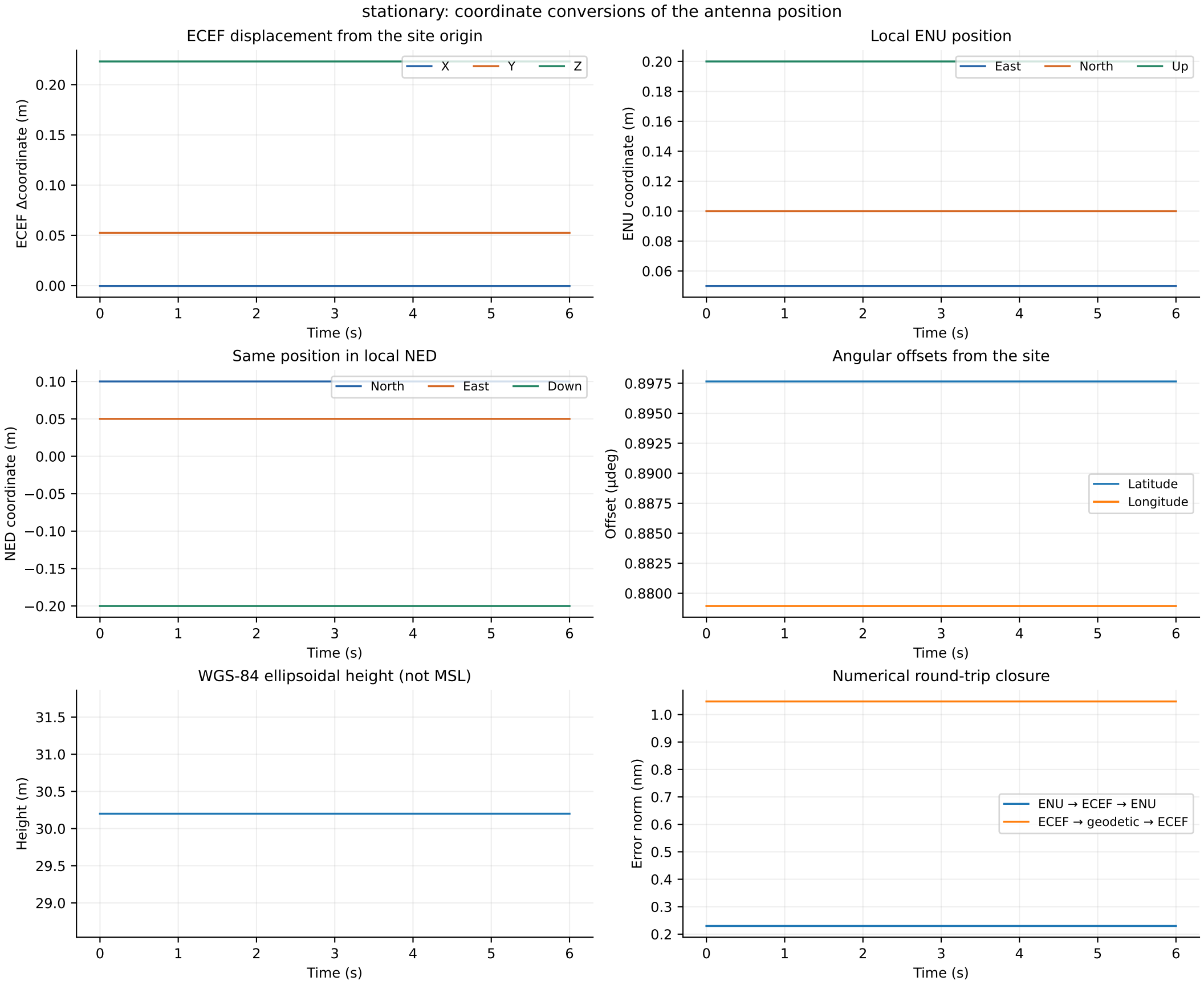

Illustrative source geometry — not real satellite ephemerides
Sky plot: elevation labels; azimuth from North
0°
0°
45°
90°
135°
180°
225°
270°
315°
30°
60°
90°
S01
S02
S03
S04
S05
S06
S07
S08
Shared direction
Receiver → source unit directions
East unit component
North unit component
Up unit component
−1.0
−0.5
0.0
0.5
1.0

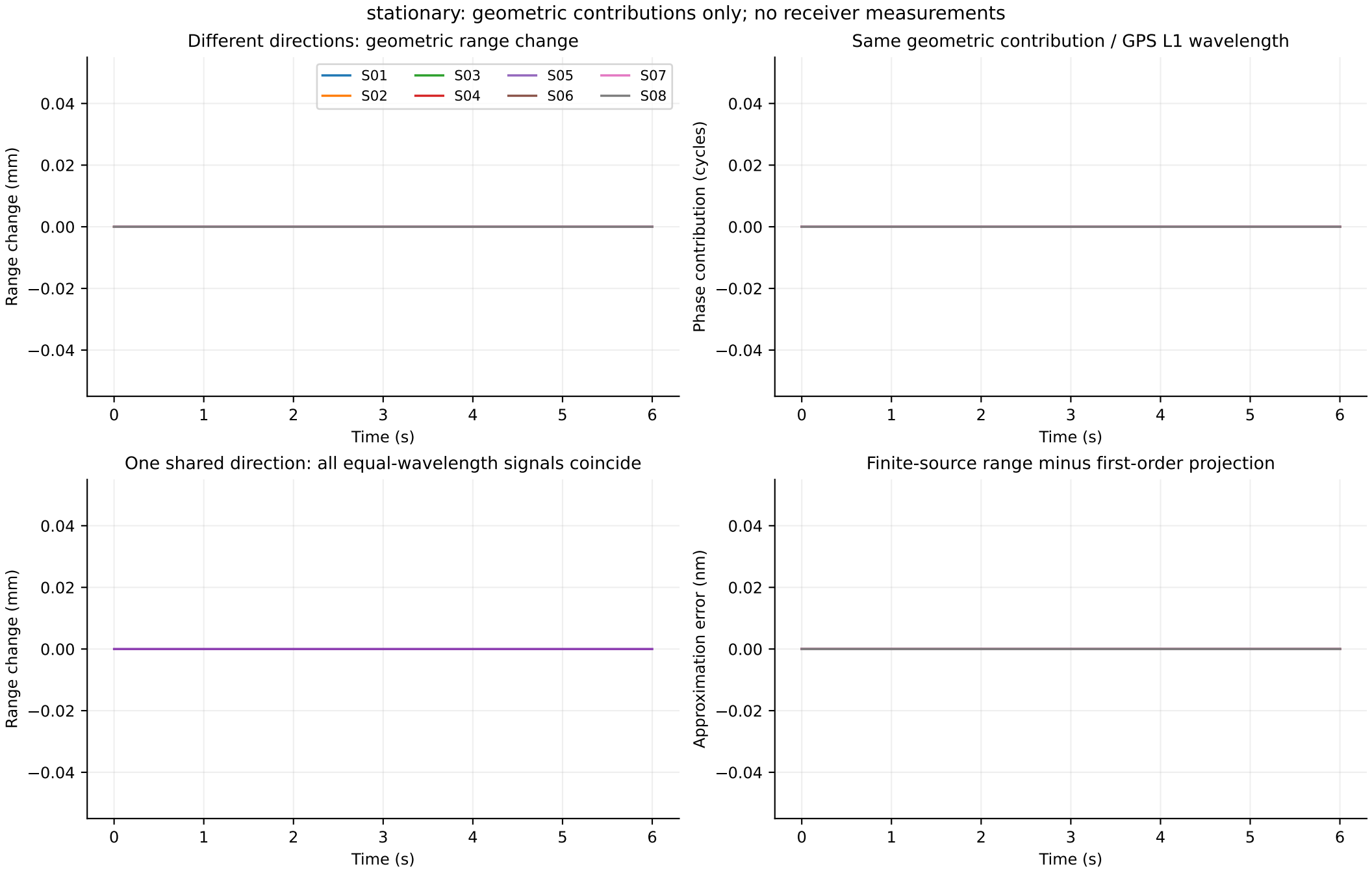

Motion controls — each panel uses equal 3D axis scales, independently fitted
stationary
constant_velocity
east_oscillation
IMU
Antenna
East (m)
North (m)
Up (m)
circle
circle_coning
three_axis
rotation_only
combined


stationary: body and antenna movement
3D path; rods connect the IMU to the antenna
IMU/body origin
Antenna
East (m)
North (m)
Up (m)
Top view (equal distance scale)
Start
East (m)
North (m)

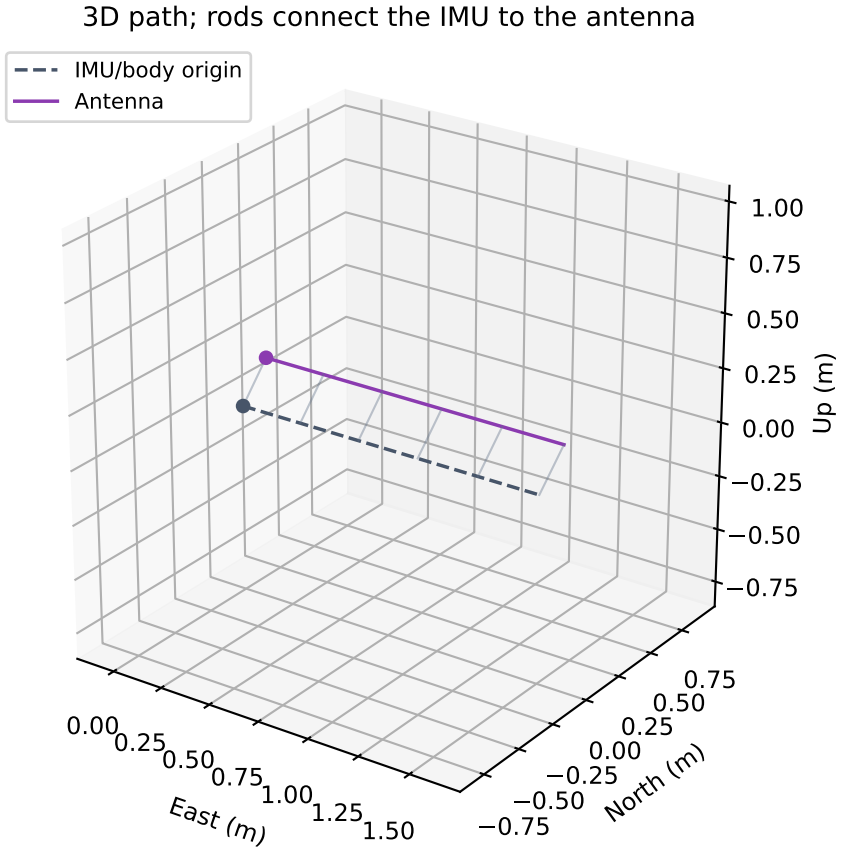

stationary: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6

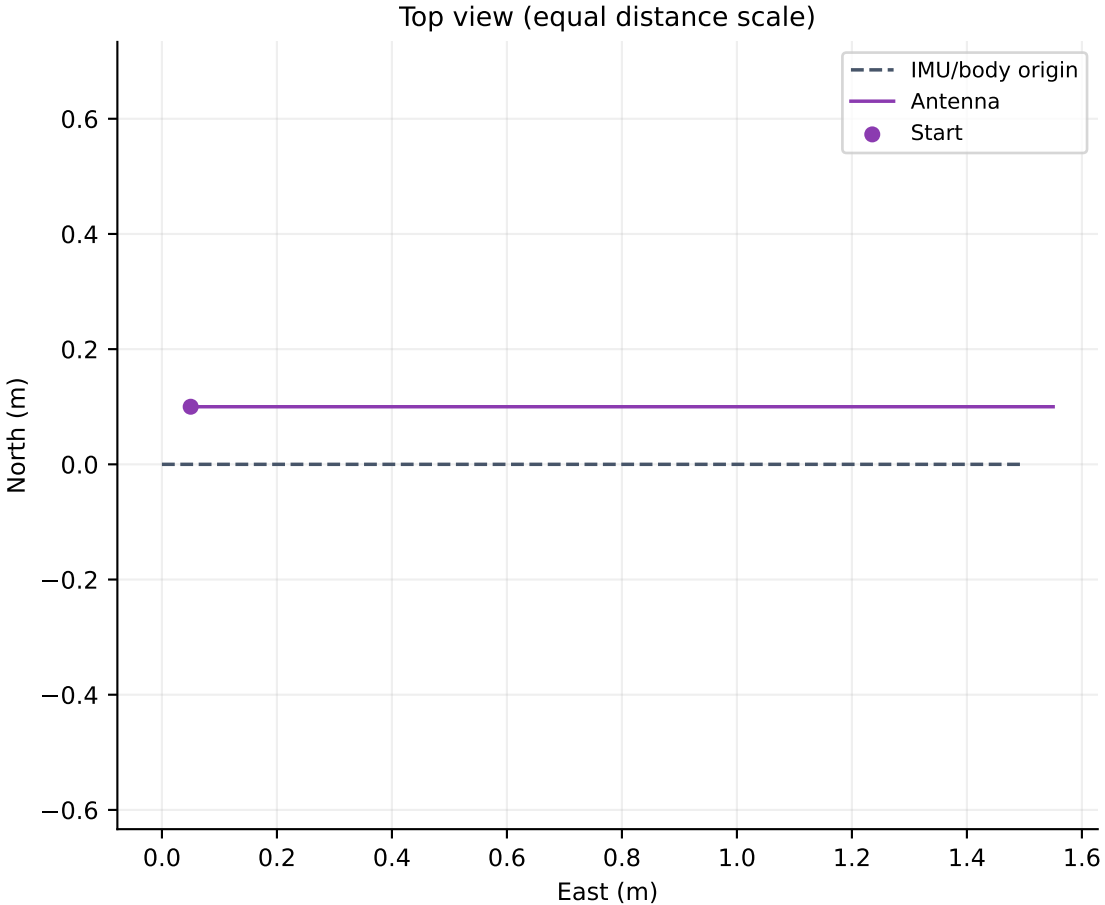

stationary: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
w
x
y
z
Quaternion component
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

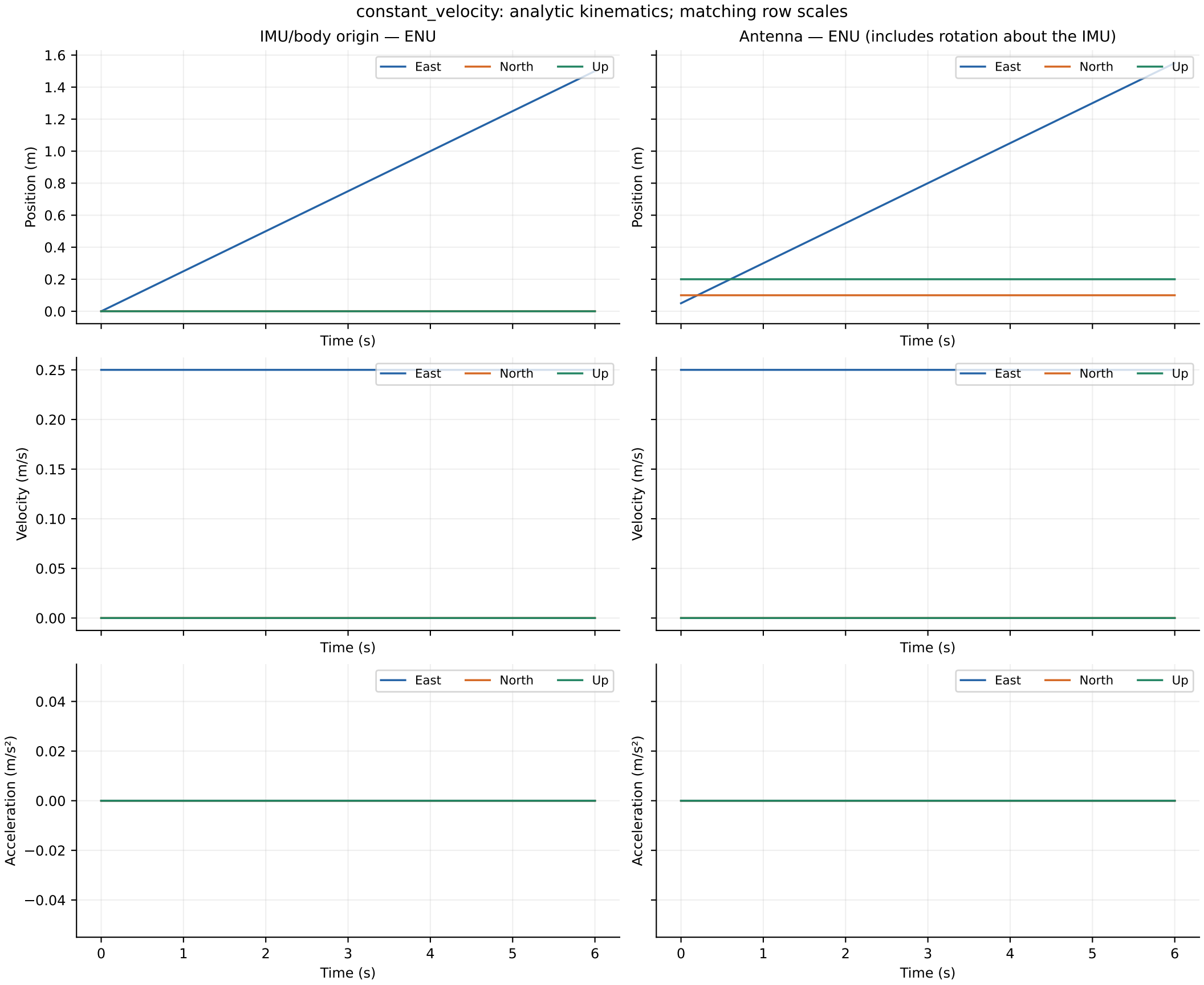

stationary: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (µdeg)
Latitude
Longitude
Time (s)
WGS-84 ellipsoidal height (not MSL)
Height (m)
Time (s)
Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)

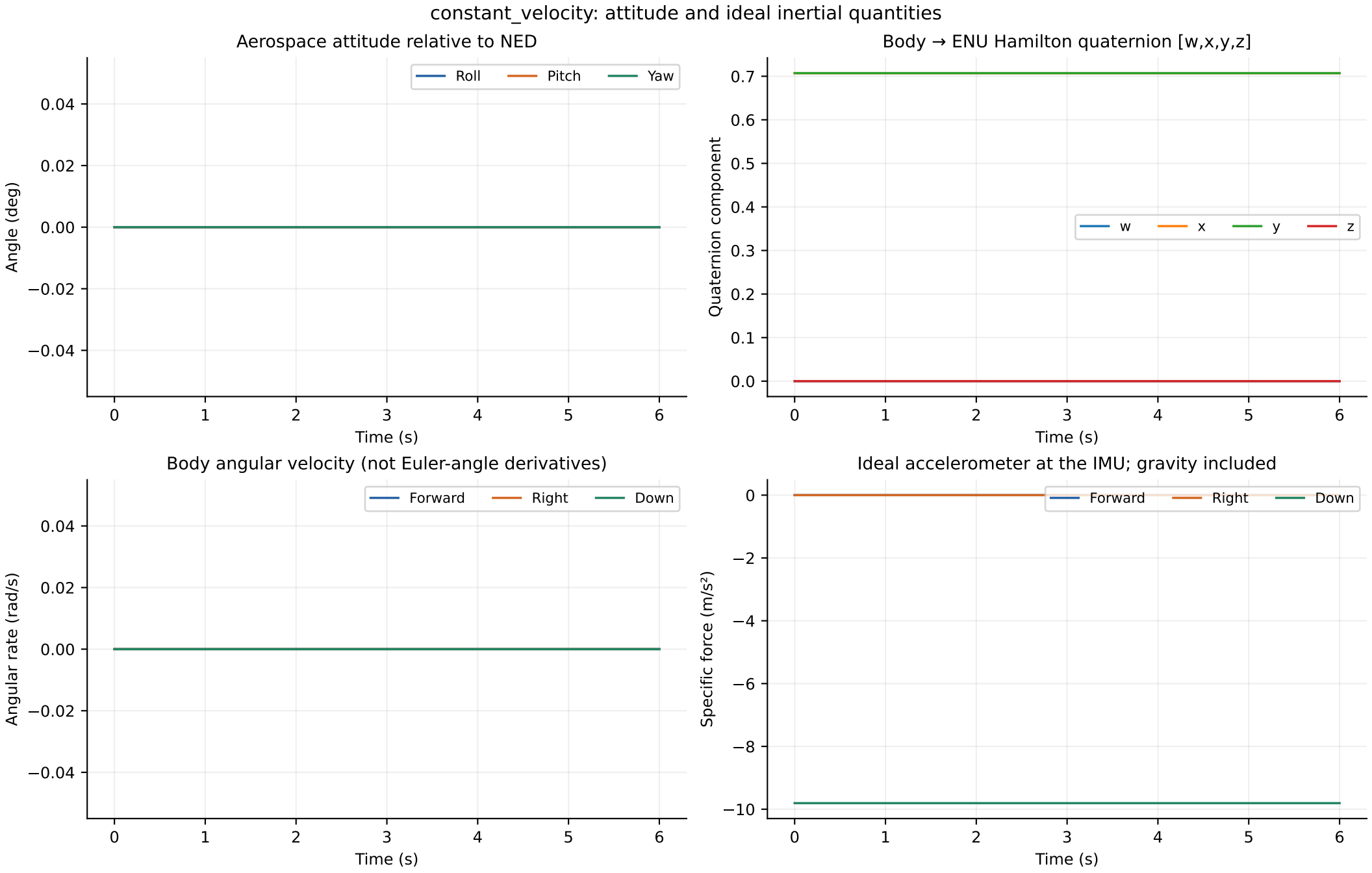

stationary: geometric contributions only; no receiver measurements
Different directions: geometric range change
Same geometric contribution / GPS L1 wavelength
One shared direction: all equal-wavelength signals coincide
Finite-source range minus first-order projection
S01
S02
S03
S04
S05
S06
S07
S08
Range change (mm)
Phase contribution (cycles)
Approximation error (nm)
Time (s)
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6

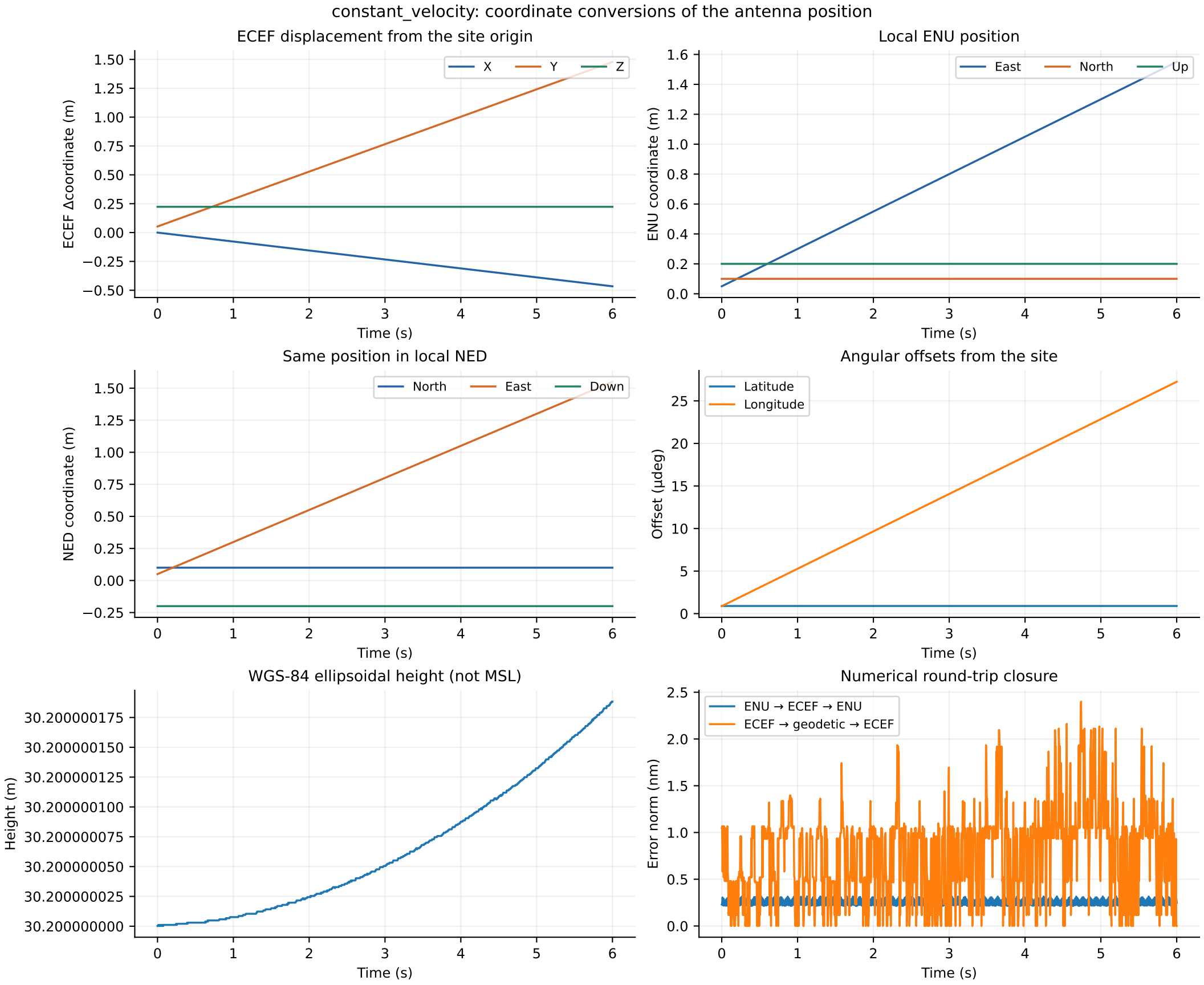
constant_velocity: body and antenna movement

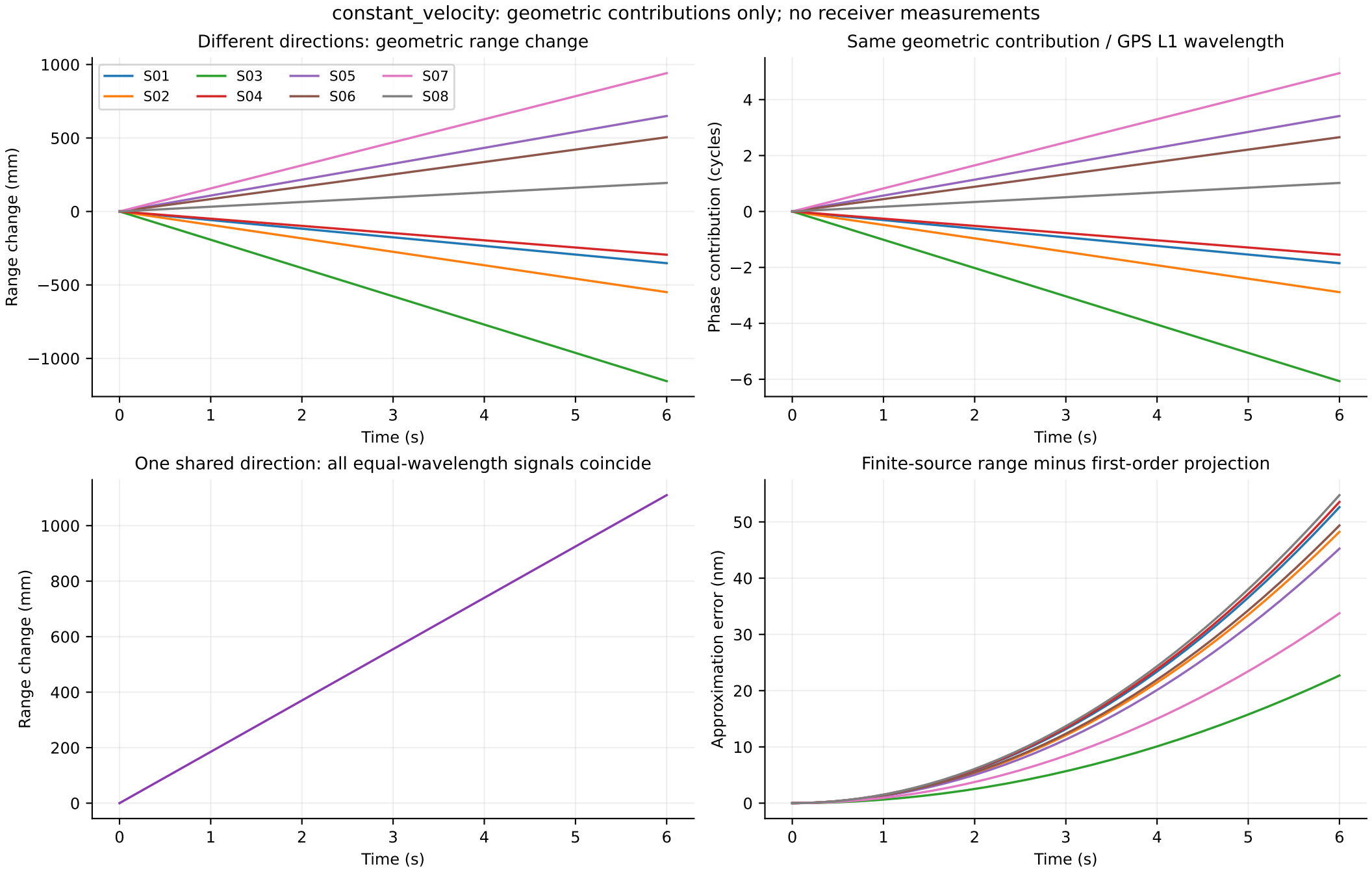


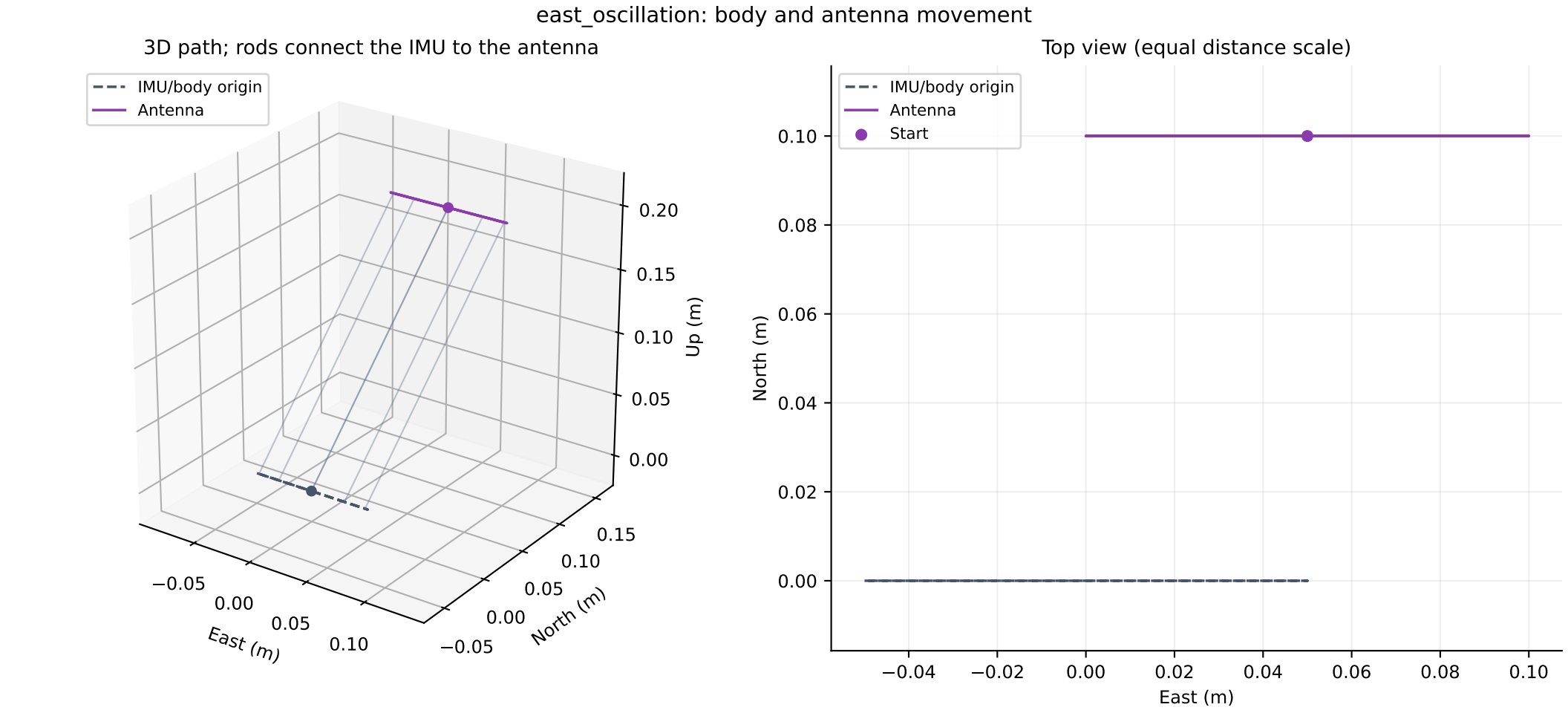


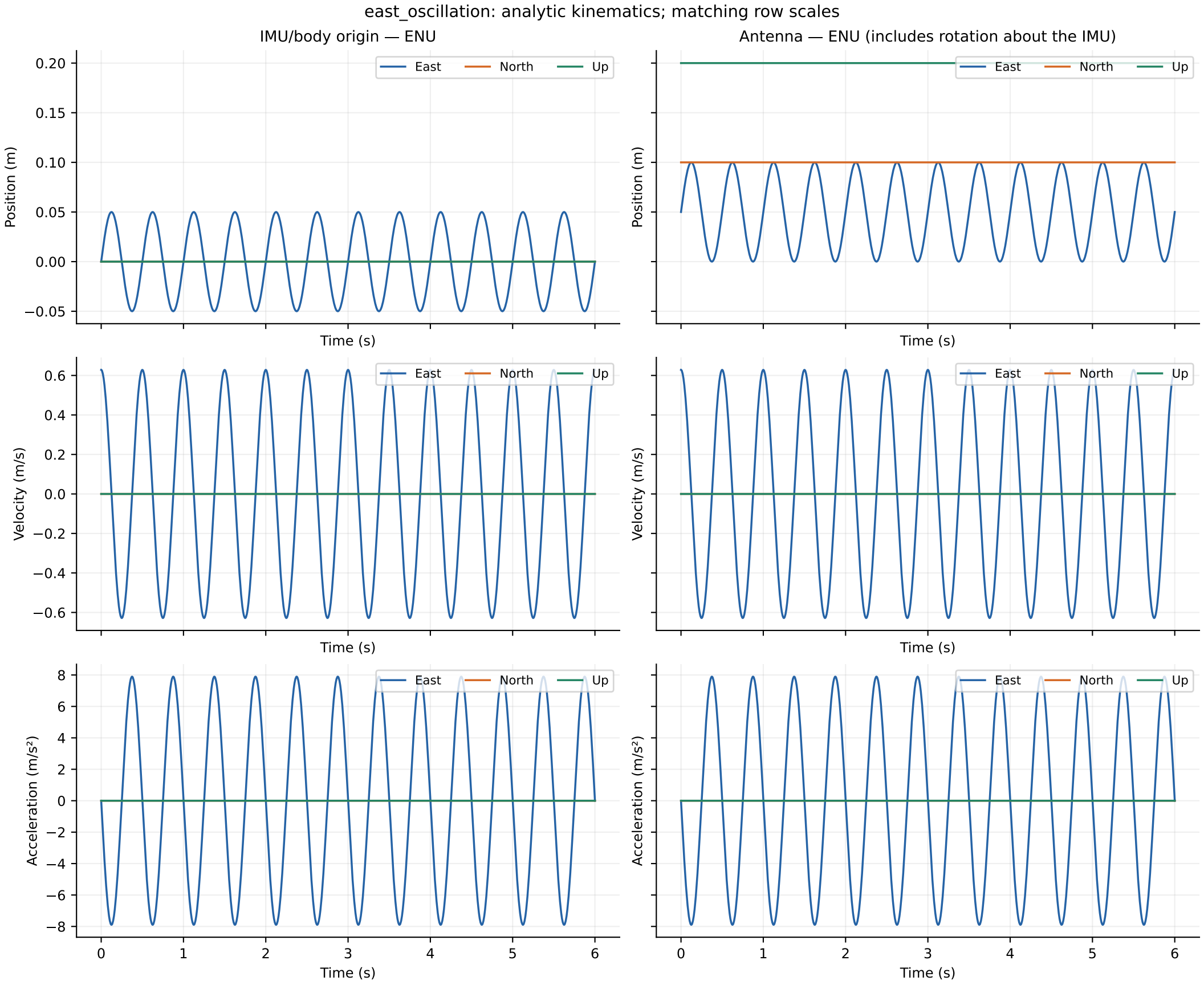

constant_velocity: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
0.00
0.05
0.10
0.15
0.20
0.25
−0.04
−0.02
0.00
0.02
0.04
0
1
2
3
4
5
6

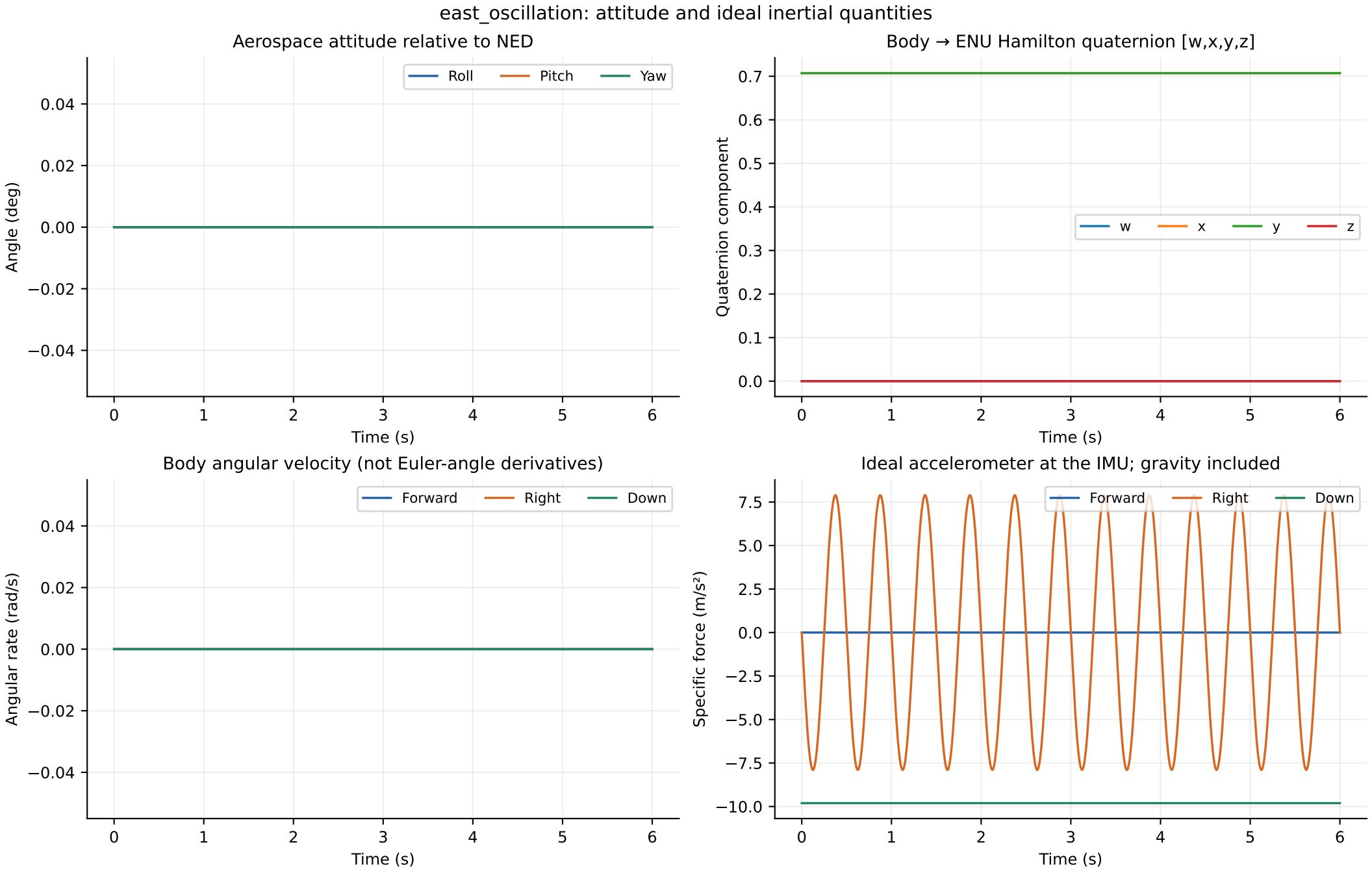

constant_velocity: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
w
x
y
z
Quaternion component
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

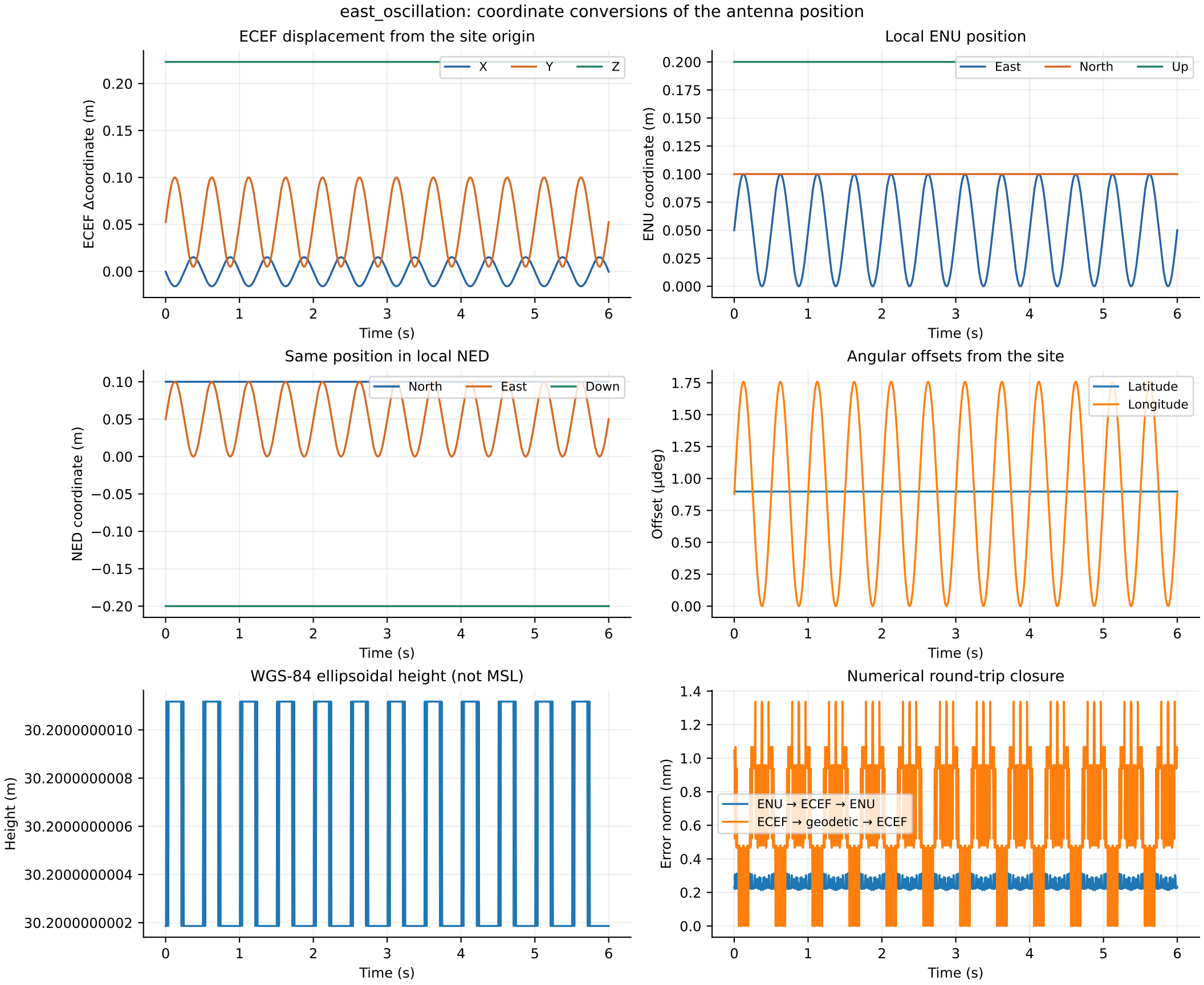

constant_velocity: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (μdeg)
Latitude
Longitude
Time (s)
WGS-84 ellipsoidal height (not MSL)
Height (m)
Time (s)
Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)

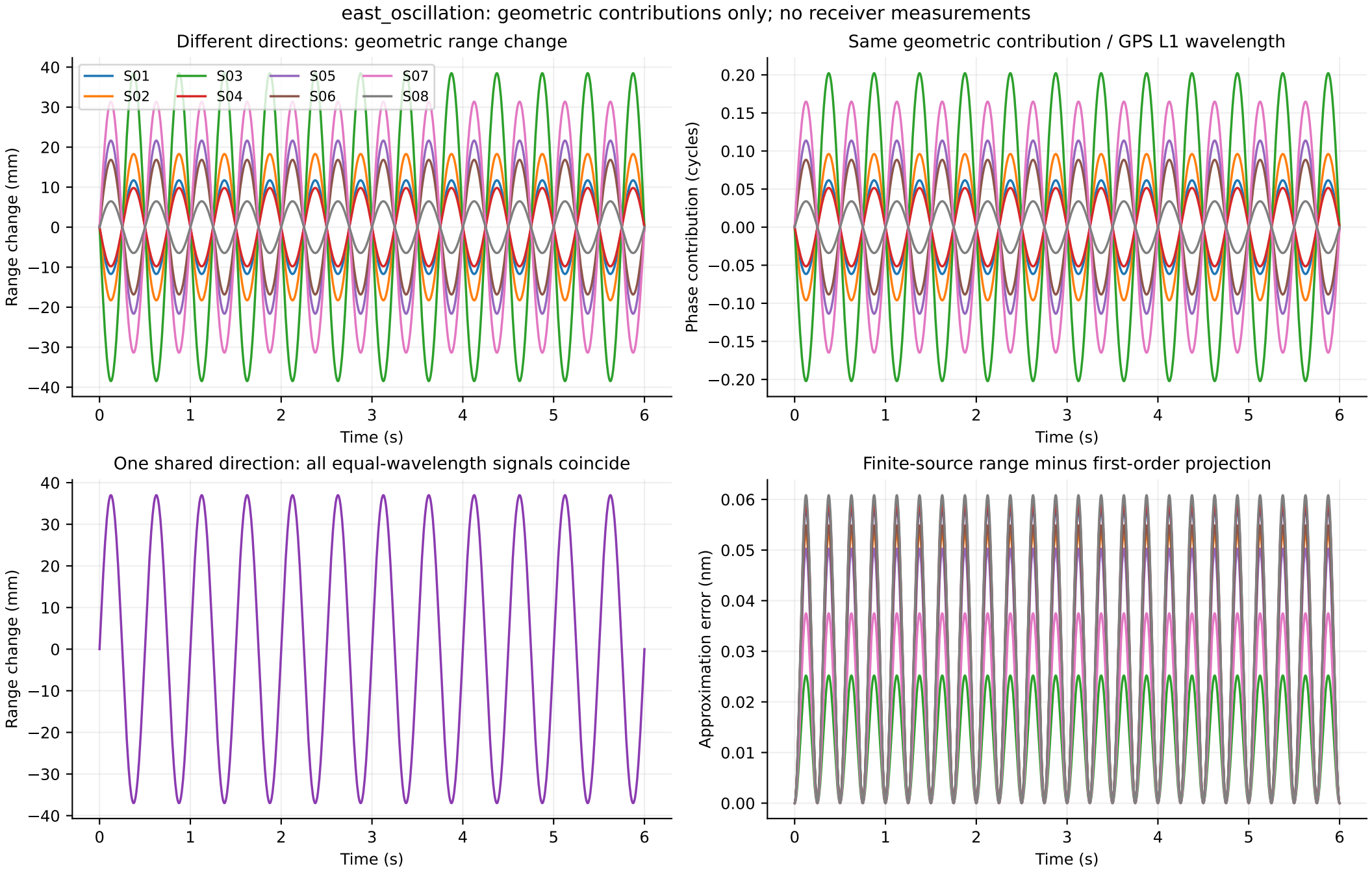

constant_velocity: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
S02
S03
S04
S05
S06
S07
S08
Same geometric contribution / GPS L1 wavelength
Phase contribution (cycles)
Time (s)
One shared direction: all equal-wavelength signals coincide
Range change (mm)
Time (s)
Finite-source range minus first-order projection
Approximation error (nm)
Time (s)

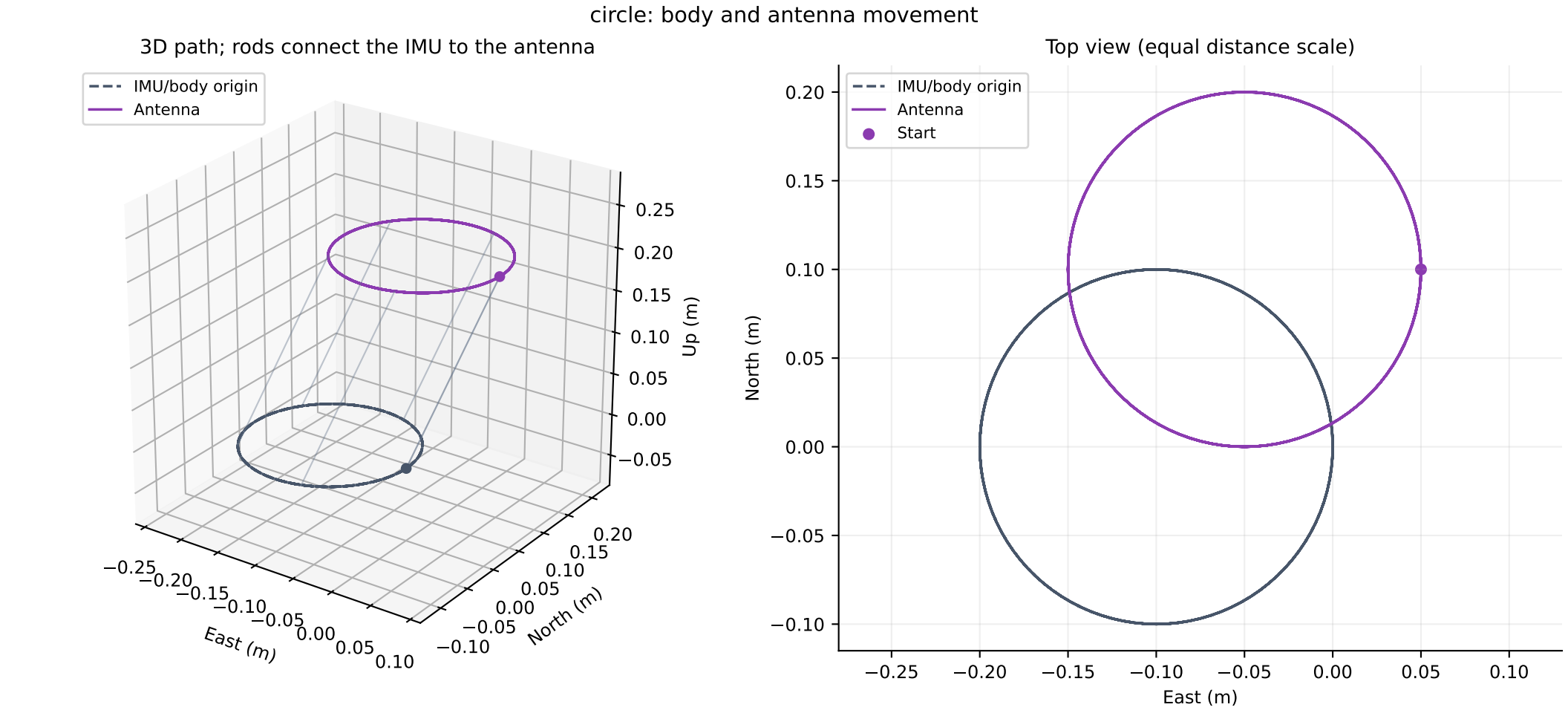

east_oscillation: body and antenna movement
3D path; rods connect the IMU to the antenna
IMU/body origin
Antenna
East (m)
North (m)
Up (m)
Top view (equal distance scale)
IMU/body origin
Antenna
Start
North (m)
East (m)

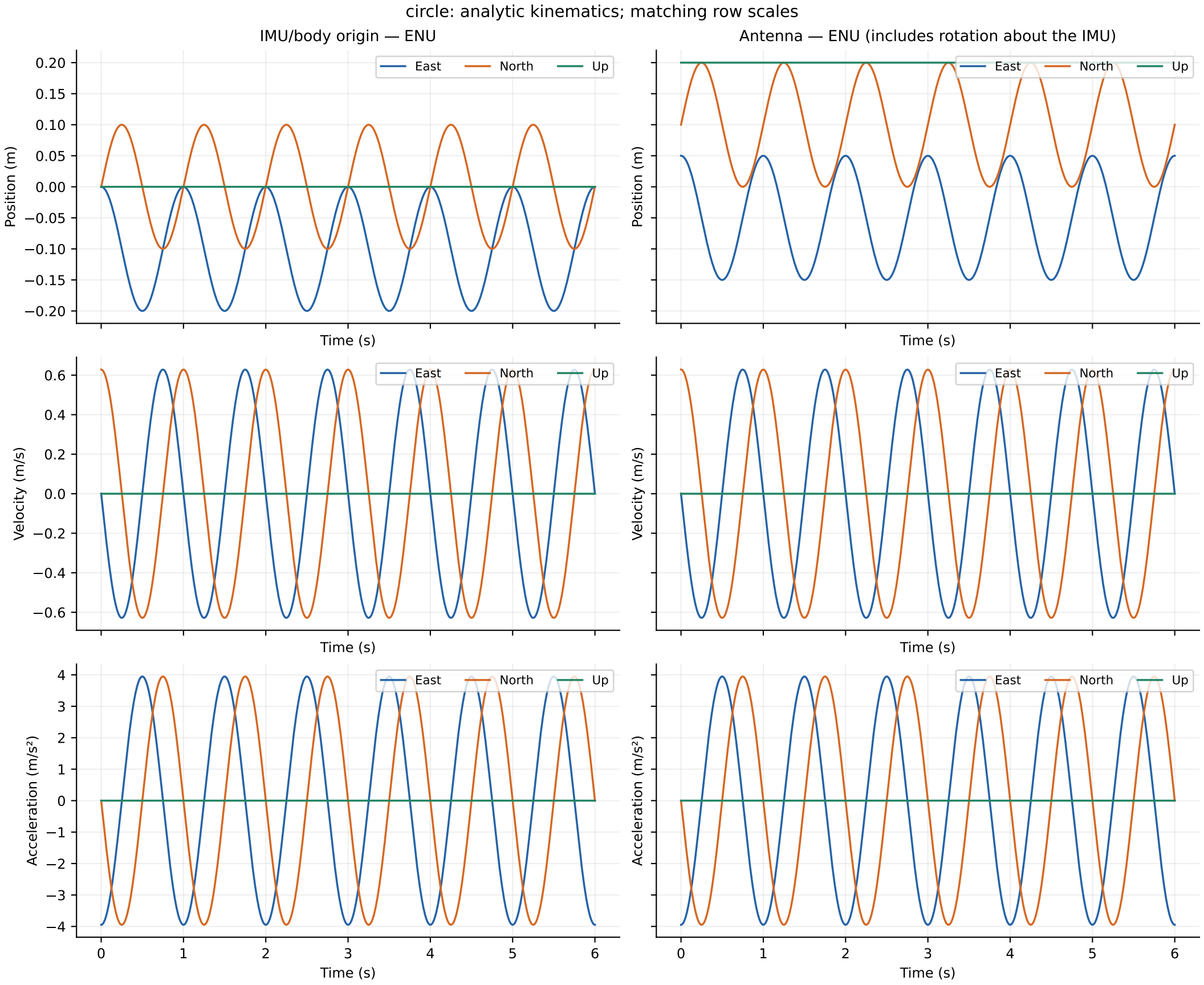

east_oscillation: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)
0.20
0.15
0.10
0.05
0.00
−0.05
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
8
6
4
2
0
−2
−4
−6
−8
0
1
2
3
4
5
6

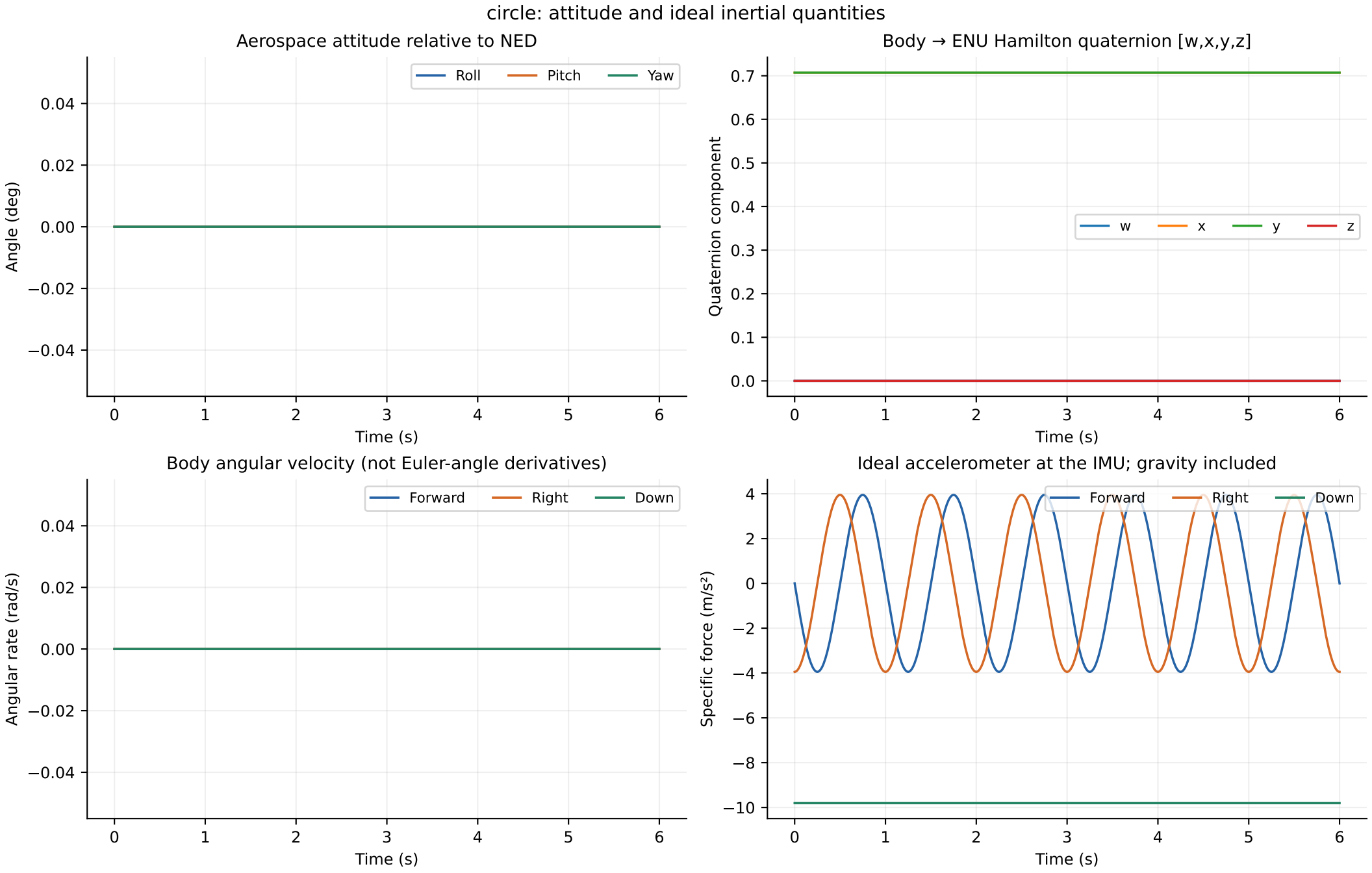

east_oscillation: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
w
x
y
z
Quaternion component
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

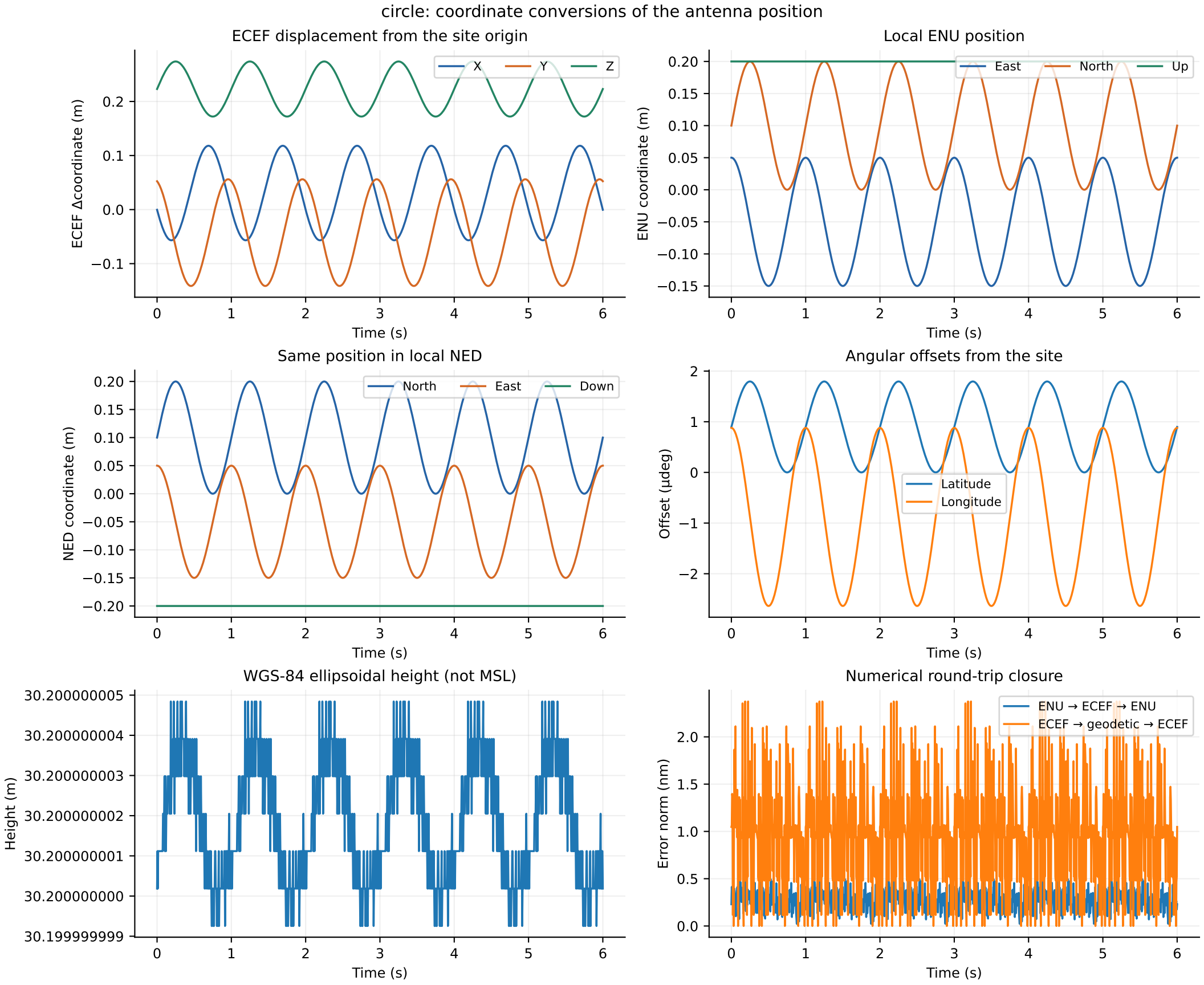

east_oscillation: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (µdeg)
Latitude
Longitude
Time (s)
WGS-84 ellipsoidal height (not MSL)
Height (m)
Time (s)
Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)

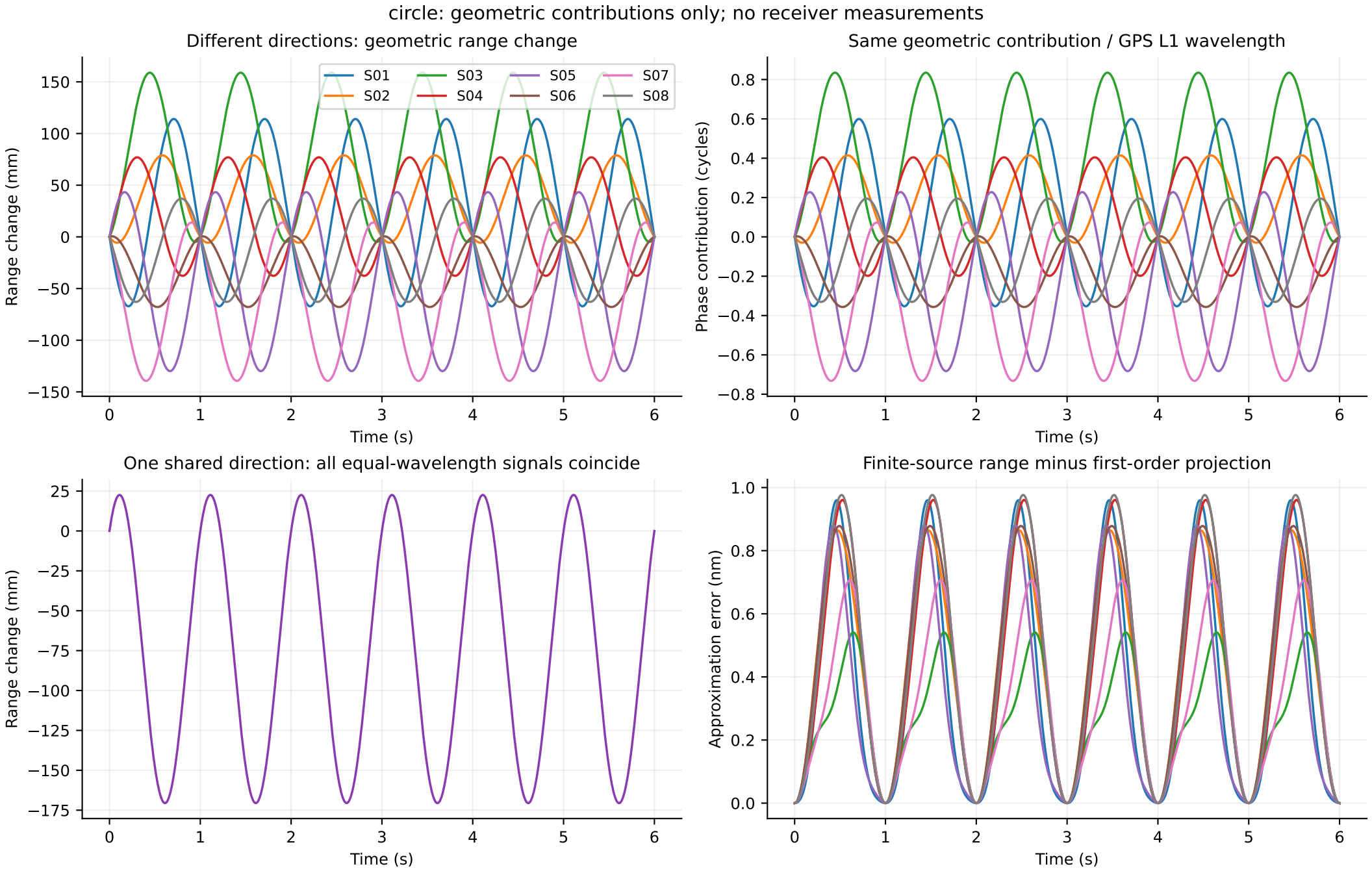

east_oscillation: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
S02
S03
S04
S05
S06
S07
S08
Same geometric contribution / GPS L1 wavelength
Phase contribution (cycles)
Time (s)
One shared direction: all equal-wavelength signals coincide
Range change (mm)
Time (s)
Finite-source range minus first-order projection
Approximation error (nm)
Time (s)

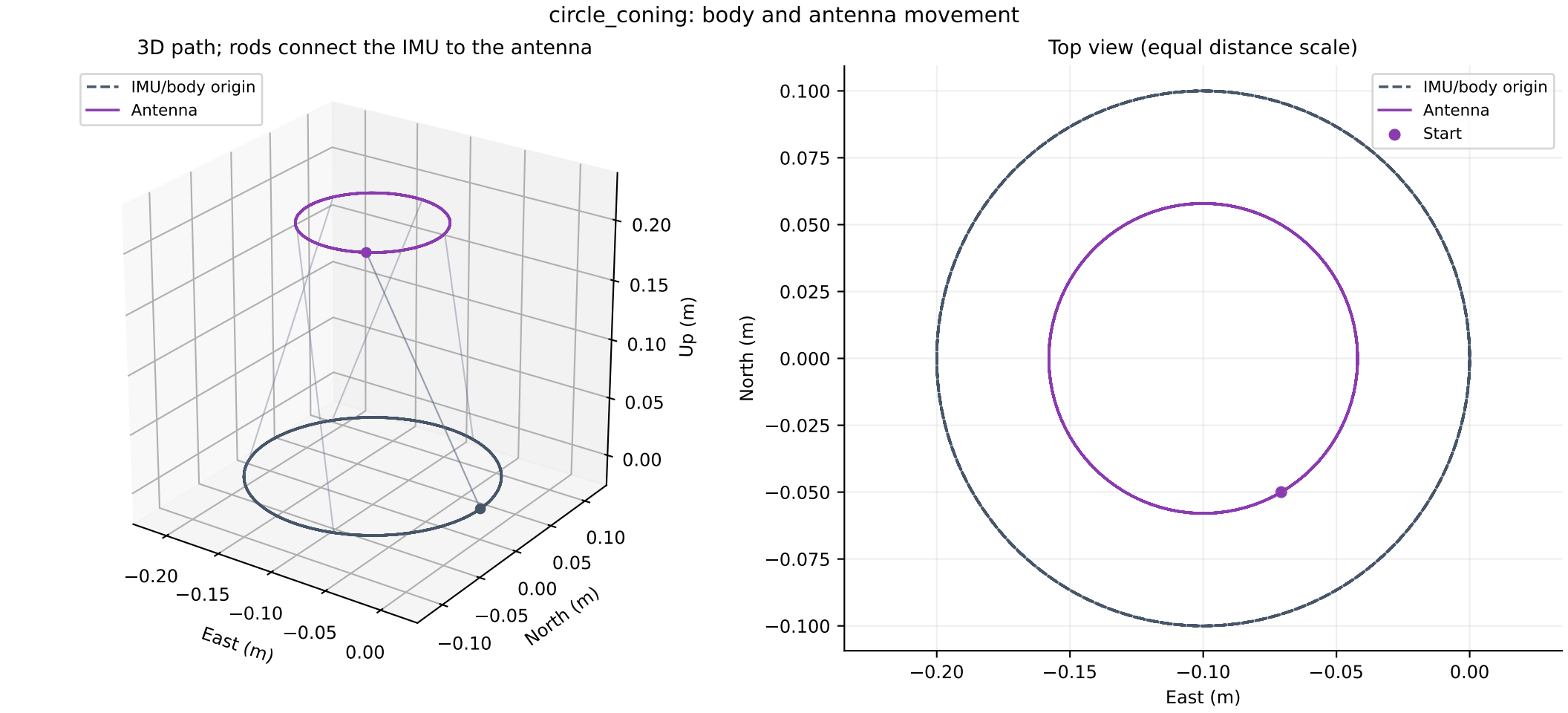

circle: body and antenna movement
3D path; rods connect the IMU to the antenna
IMU/body origin
Antenna
East (m)
North (m)
Up (m)
Top view (equal distance scale)
IMU/body origin
Antenna
Start
North (m)
East (m)

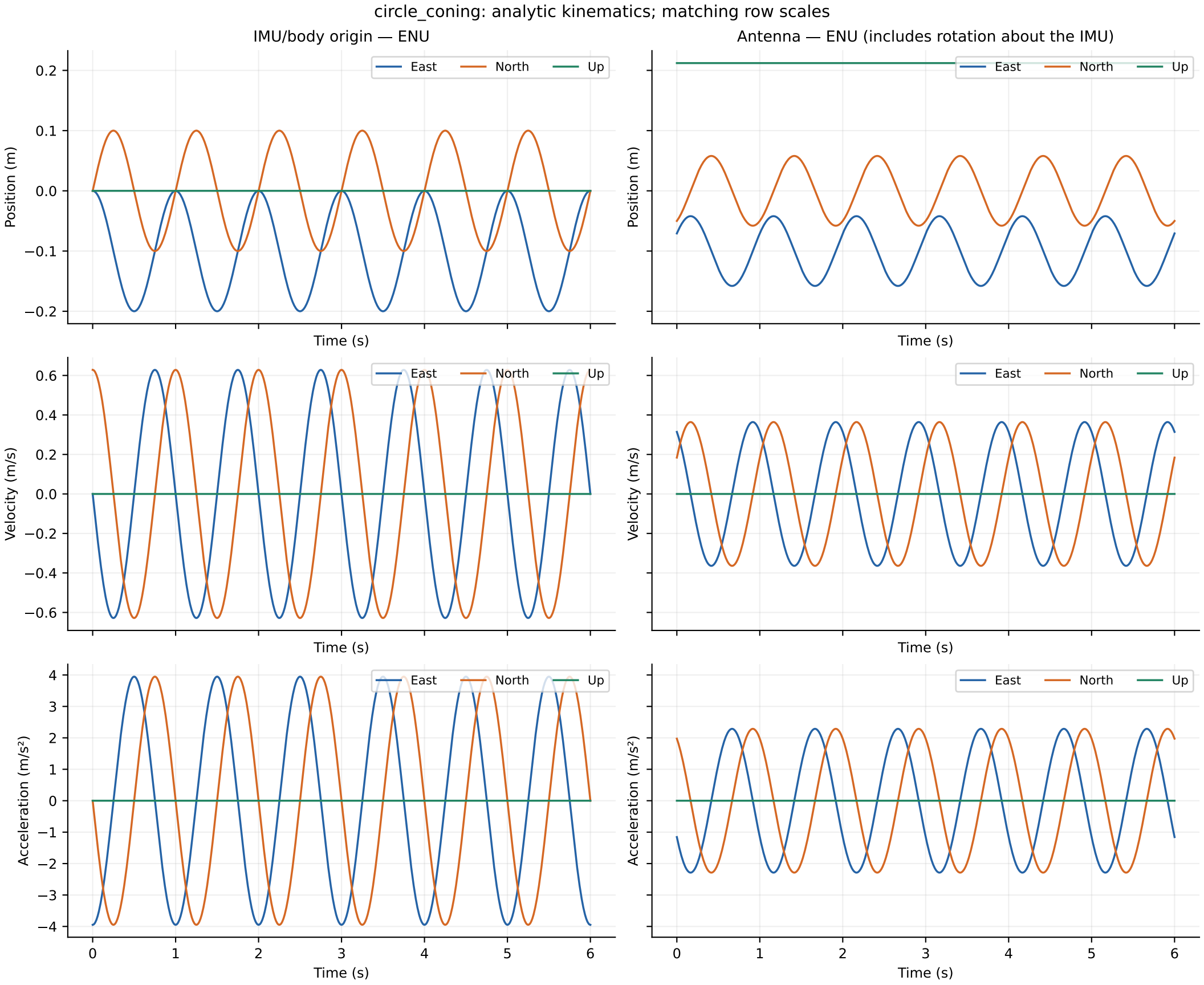

circle: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)

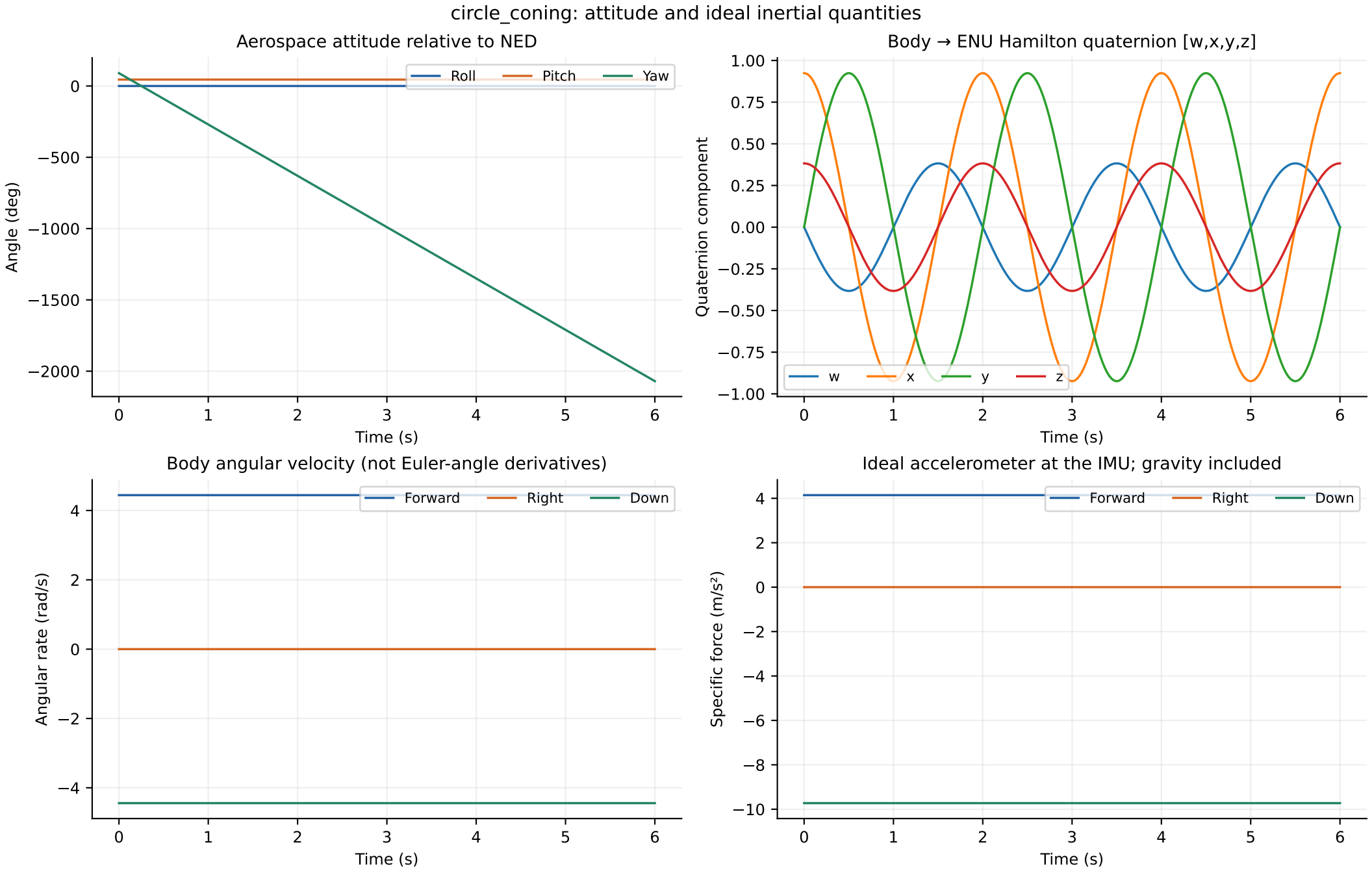

circle: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
w
x
y
z
Quaternion component
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

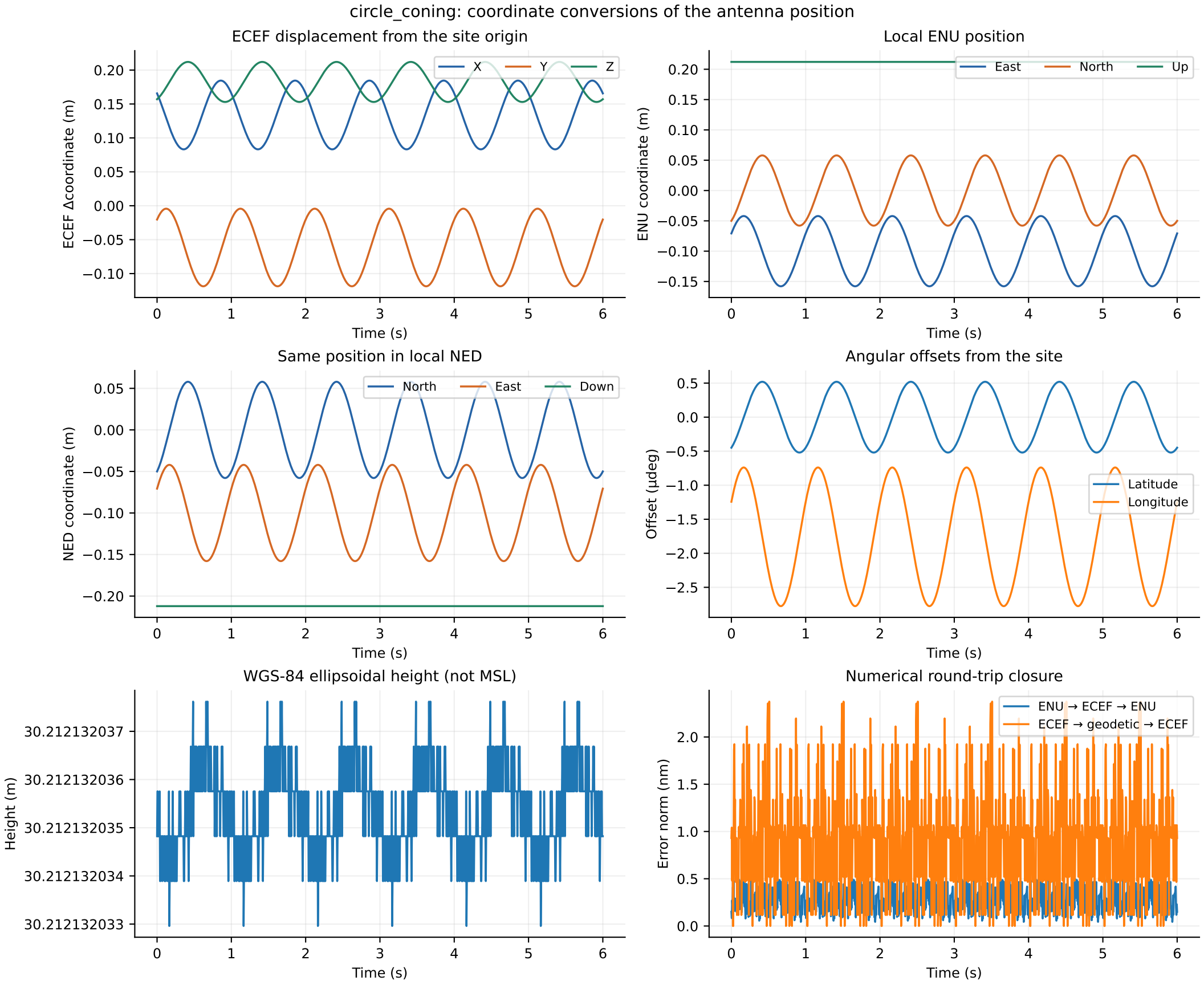

circle: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (μdeg)
Latitude
Longitude
Time (s)
WGS-84 ellipsoidal height (not MSL)
Height (m)
Time (s)
Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)

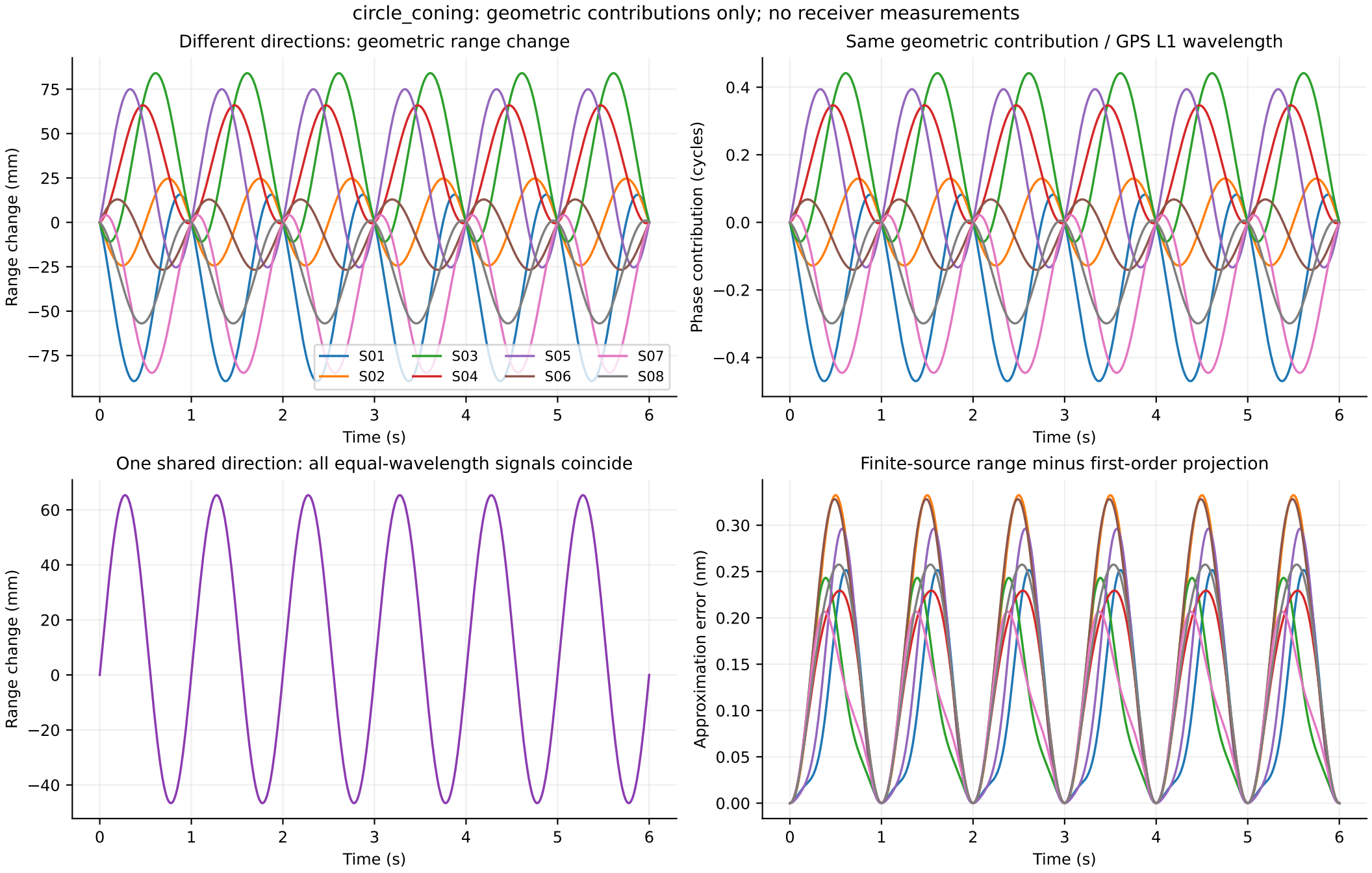

circle: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
S02
S03
S04
S05
S06
S07
S08
Same geometric contribution / GPS L1 wavelength
Phase contribution (cycles)
Time (s)
One shared direction: all equal-wavelength signals coincide
Range change (mm)
Time (s)
Finite-source range minus first-order projection
Approximation error (nm)
Time (s)

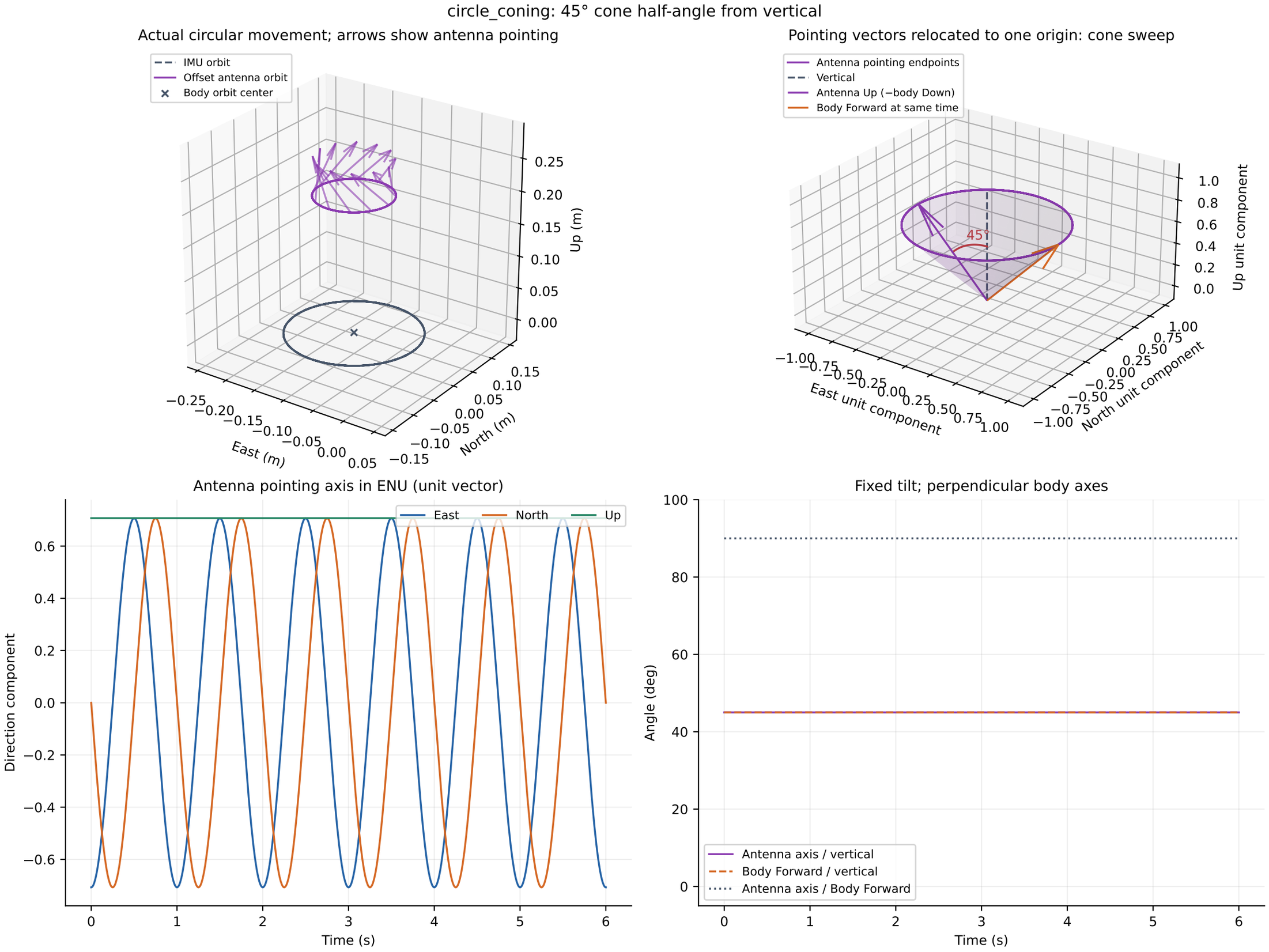

circle_coning: body and antenna movement
3D path; rods connect the IMU to the antenna
IMU/body origin
Antenna
East (m)
North (m)
Up (m)
Top view (equal distance scale)
IMU/body origin
Antenna
Start
North (m)
East (m)


circle_coning: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)


circle_coning: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
Quaternion component
w
x
y
z
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

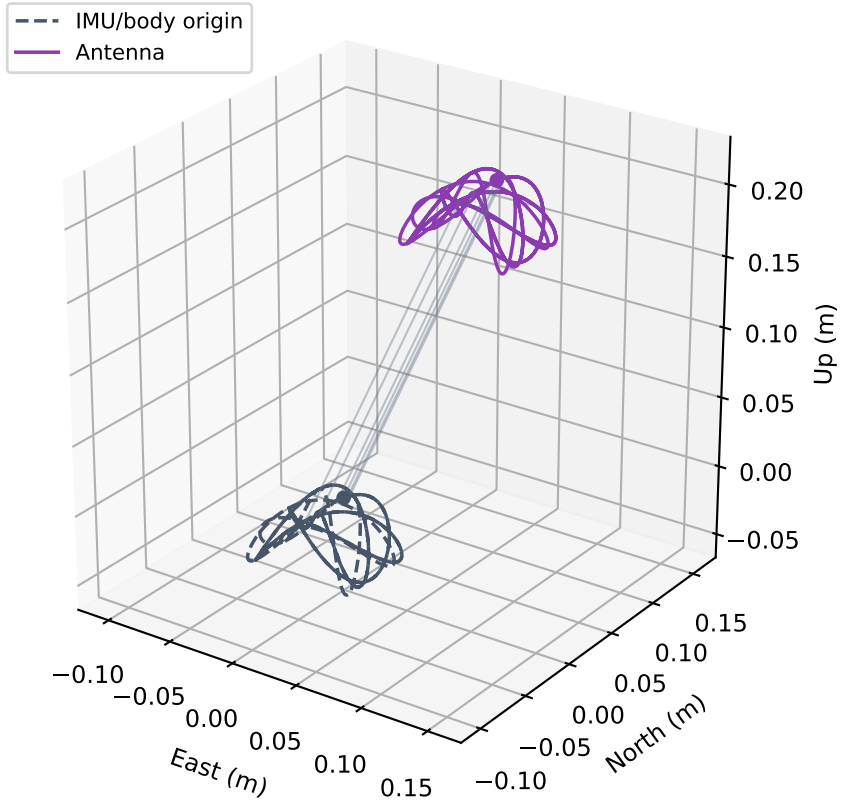

circle_coning: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (μdeg)
Latitude
Longitude
Time (s)
WGS-84 ellipsoidal height (not MSL)
Height (m)
Time (s)
Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)


circle_coning: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
S02
S03
S04
S05
S06
S07
S08
Same geometric contribution / GPS L1 wavelength
Phase contribution (cycles)
Time (s)
One shared direction: all equal-wavelength signals coincide
Range change (mm)
Time (s)
Finite-source range minus first-order projection
Approximation error (nm)
Time (s)

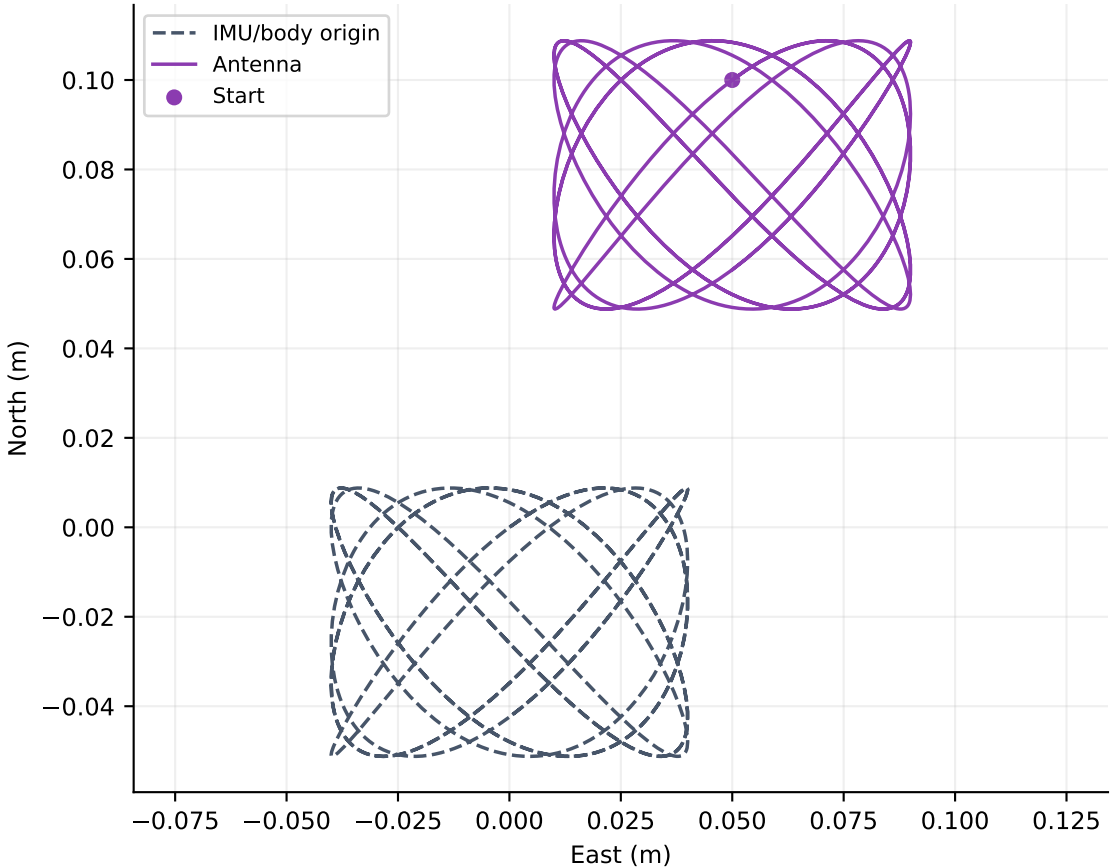

circle_coning: 45° cone half-angle from vertical
Actual circular movement; arrows show antenna pointing
IMU orbit
Offset antenna orbit
Body orbit center
East (m)
North (m)
Up (m)
Pointing vectors relocated to one origin: cone sweep
Antenna pointing endpoints
Vertical
Antenna Up (−body Down)
Body Forward at same time
45°
East unit component
North unit component
Up unit component
Antenna pointing axis in ENU (unit vector)
East
North
Up
Direction component
Time (s)
Fixed tilt; perpendicular body axes
Antenna axis / vertical
Body Forward / vertical
Antenna axis / Body Forward
Angle (deg)
Time (s)

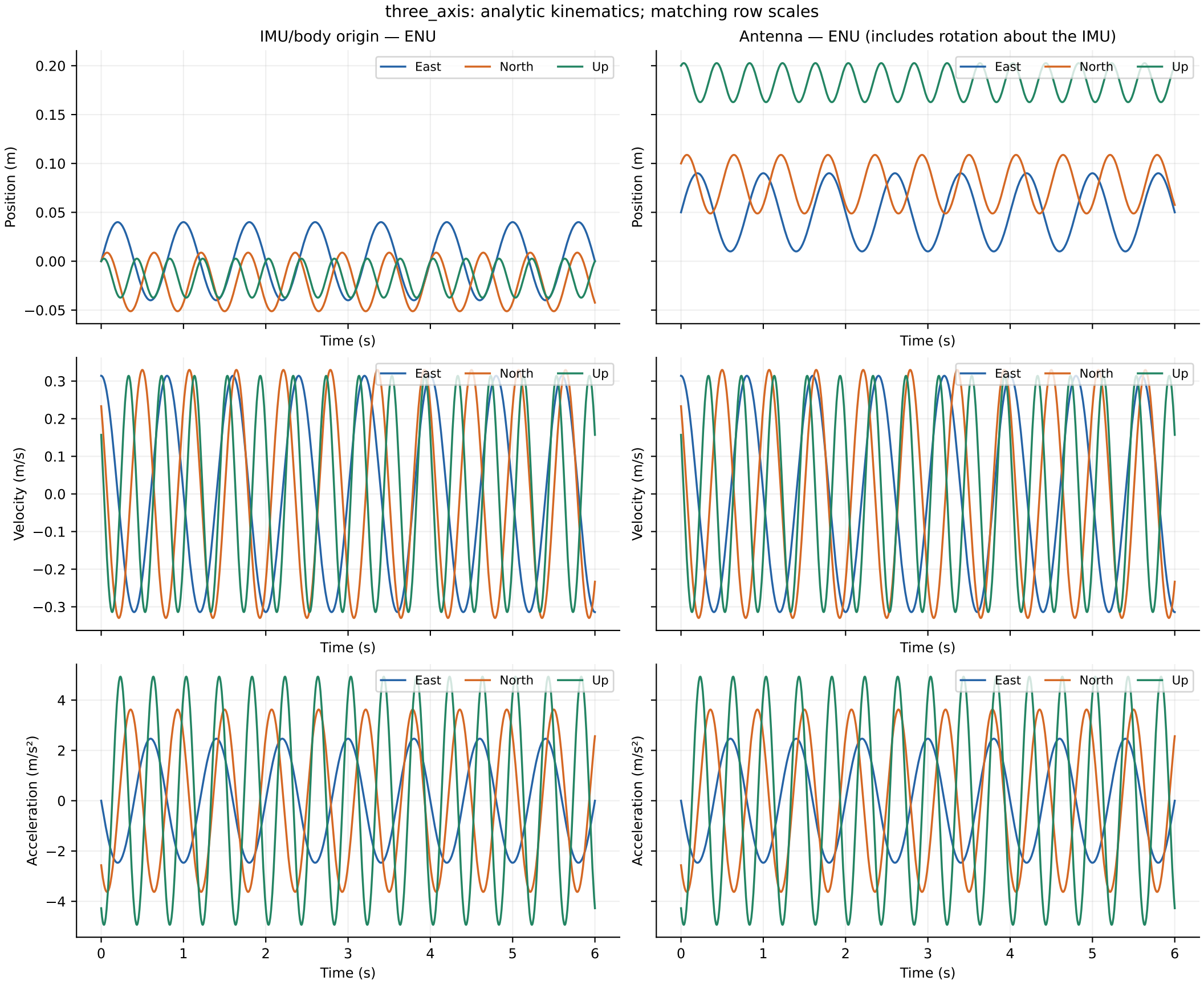
three_axis: body and antenna movement

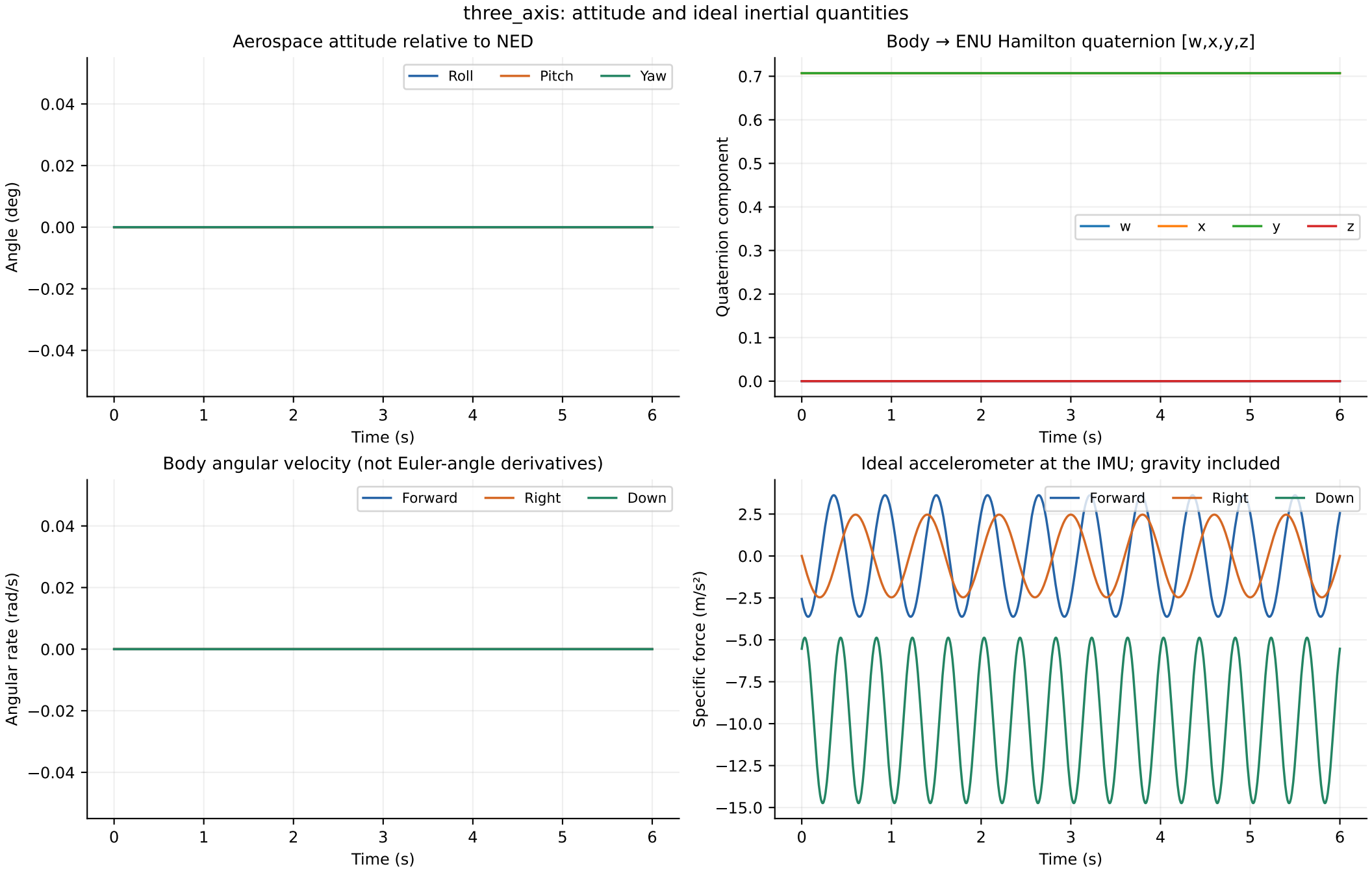
3D path; rods connect the IMU to the antenna



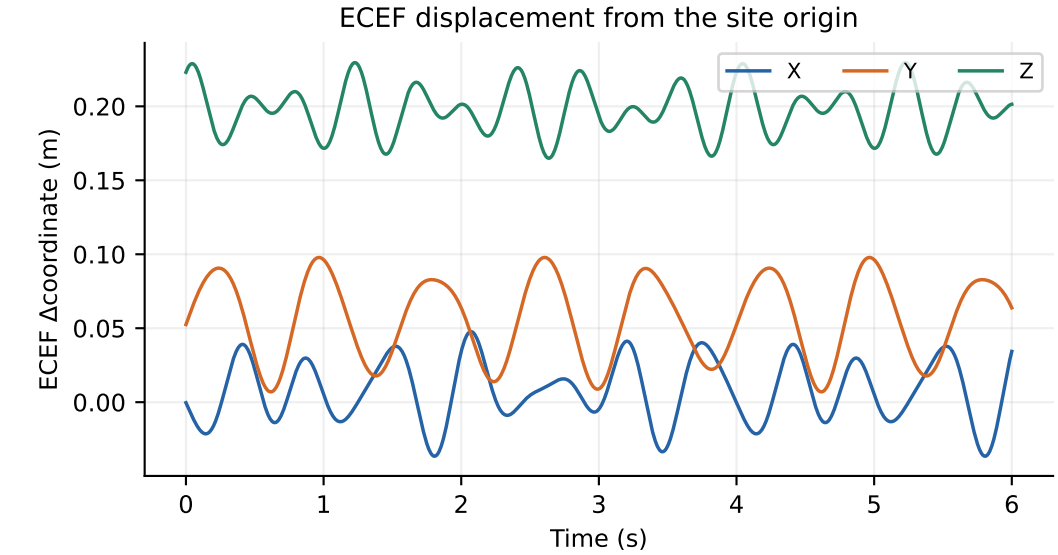
Top view (equal distance scale)

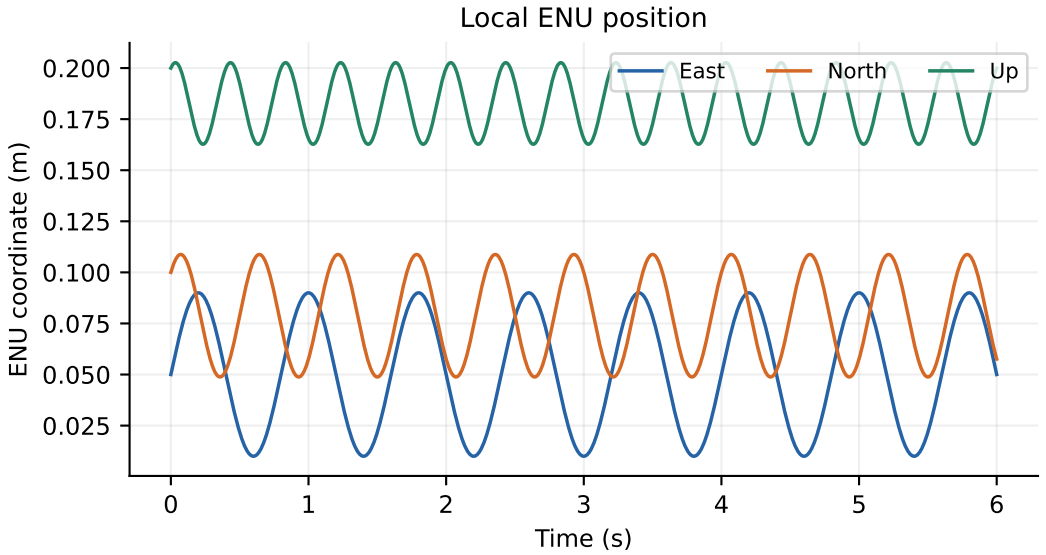


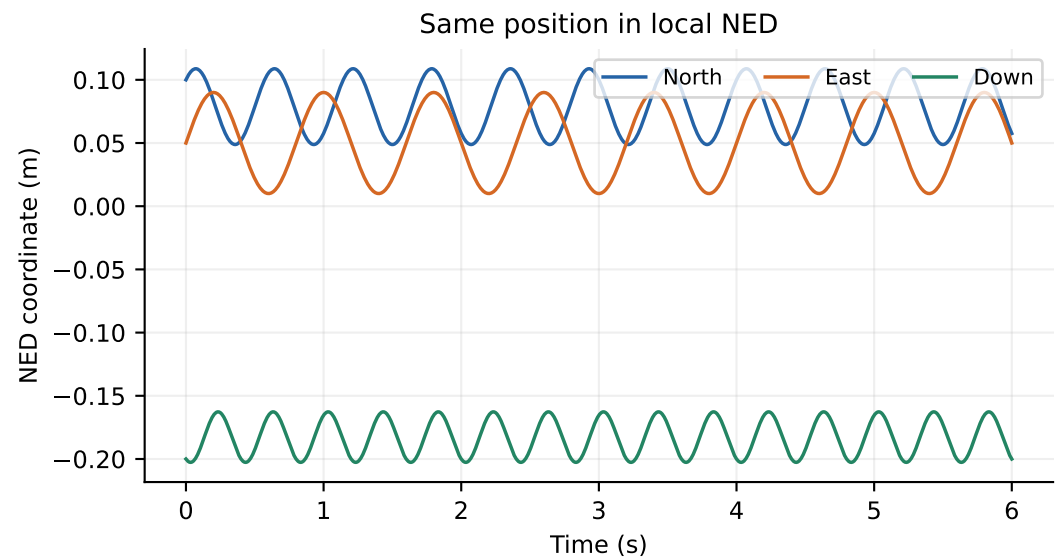

three_axis: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)
0.20
0.15
0.10
0.05
0.00
−0.05
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
4
2
0
−2
−4
0
1
2
3
4
5
6

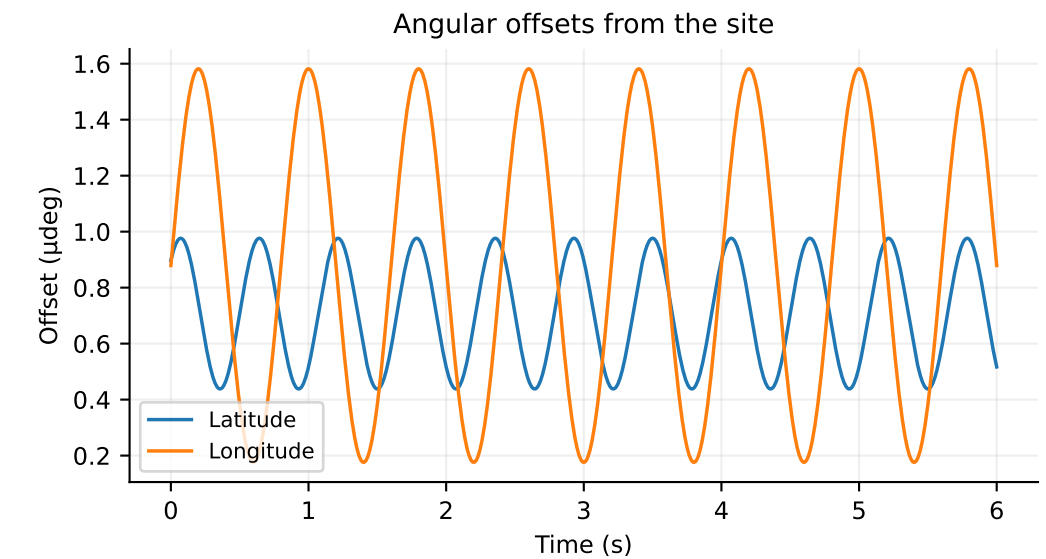

three_axis: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
w
x
y
z
Quaternion component
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

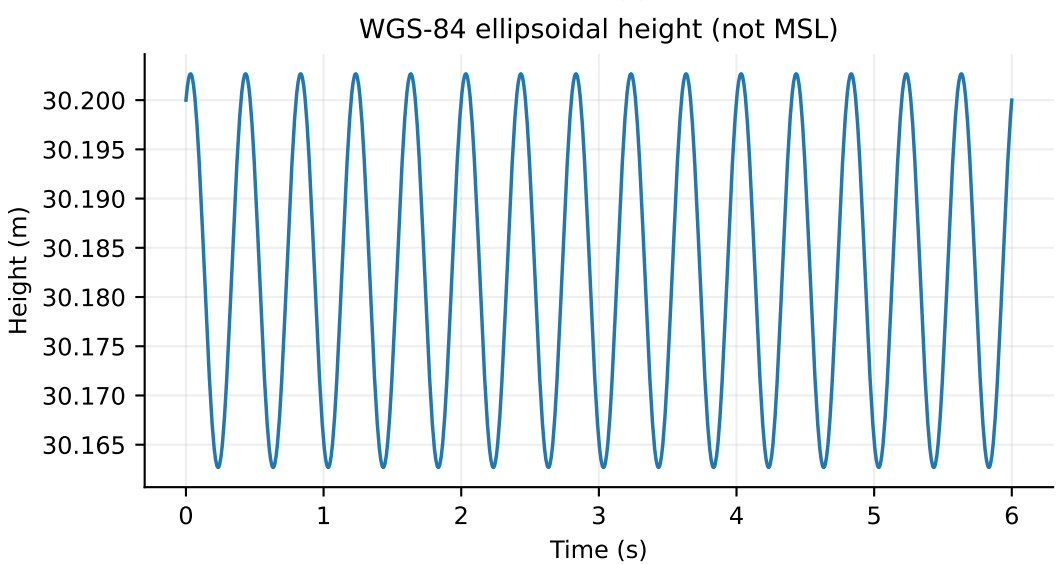
three_axis: coordinate conversions of the antenna position

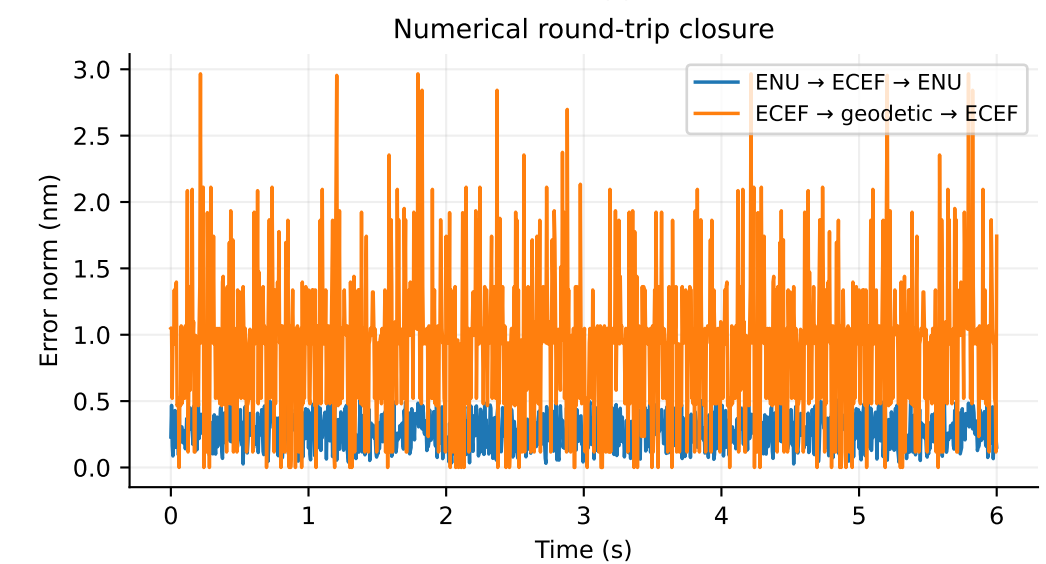


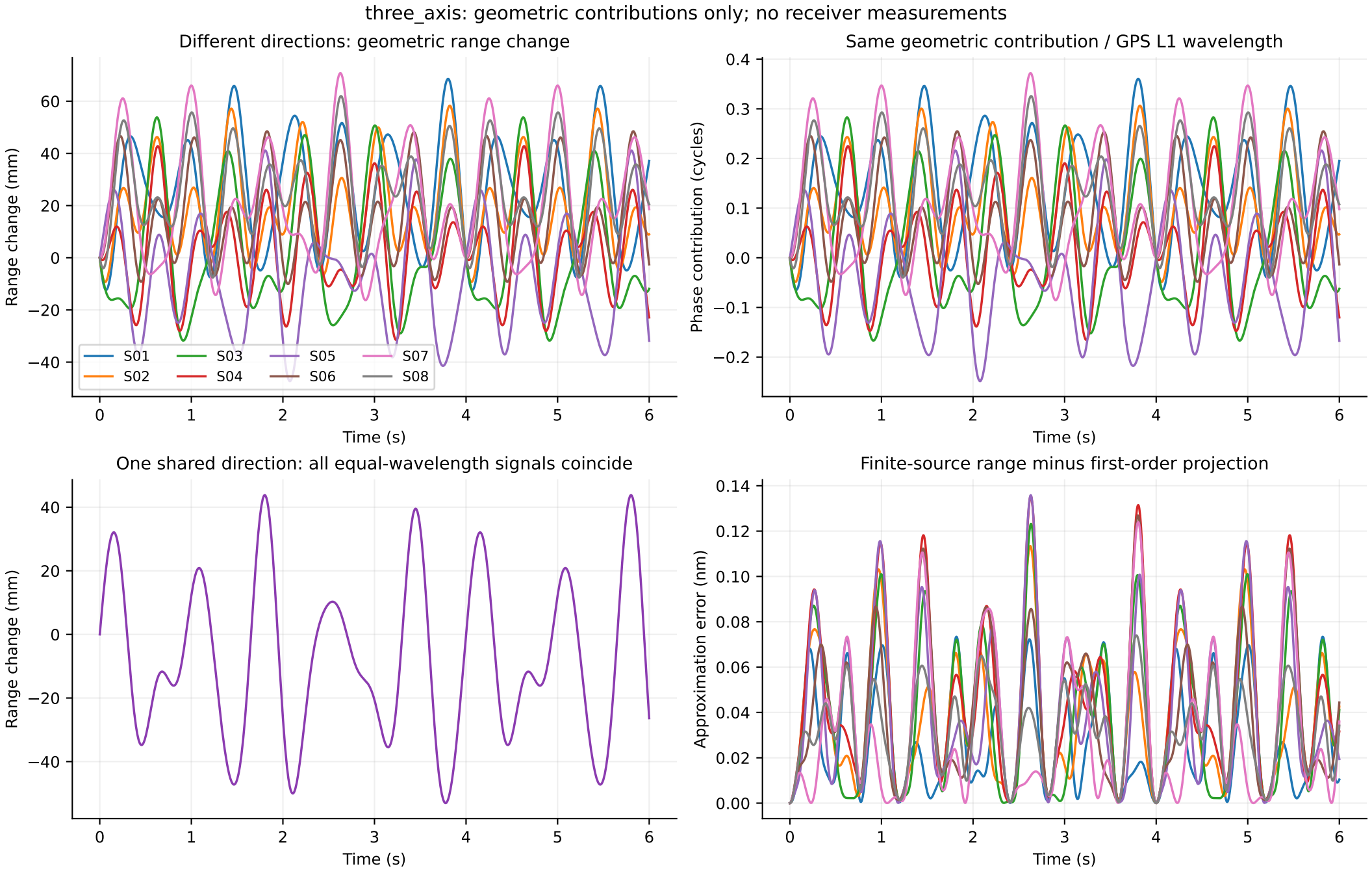


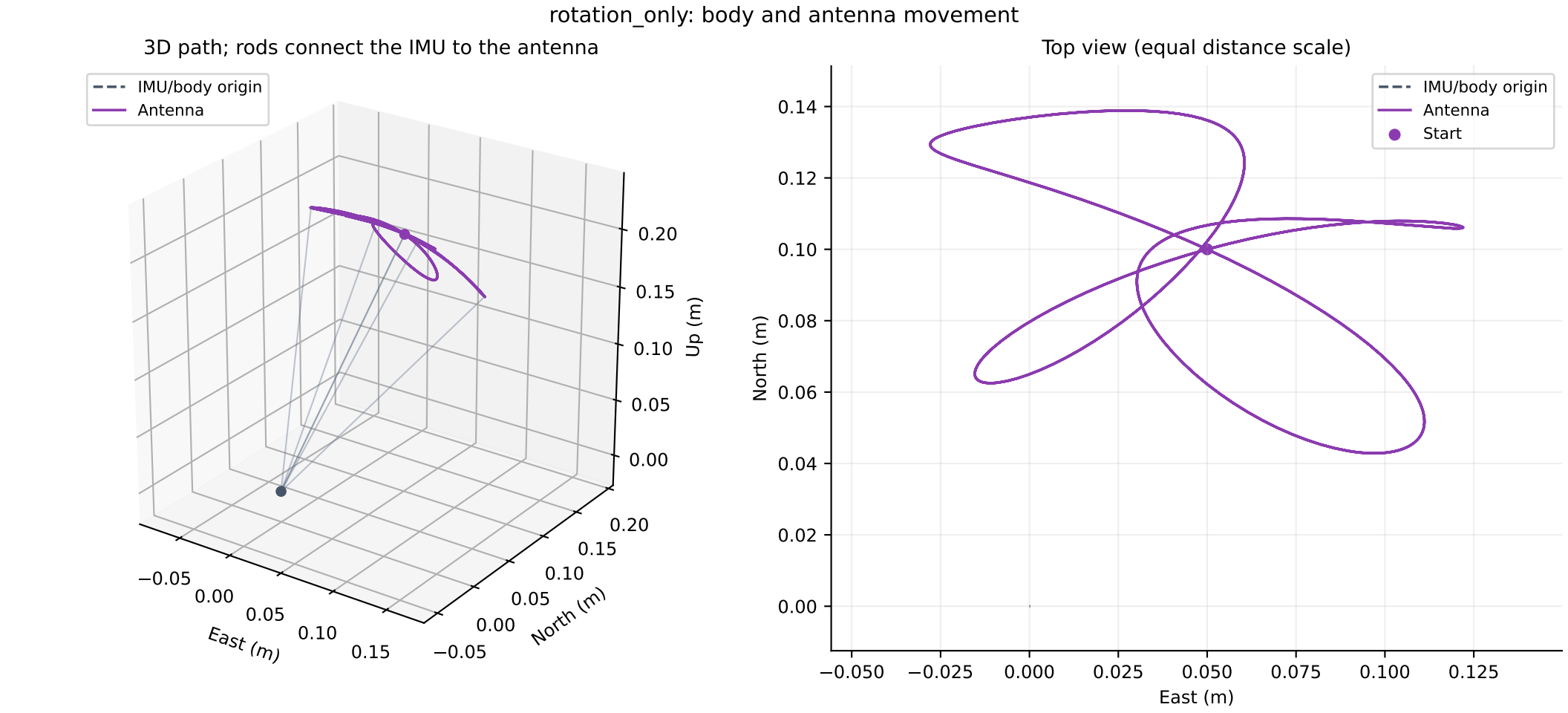


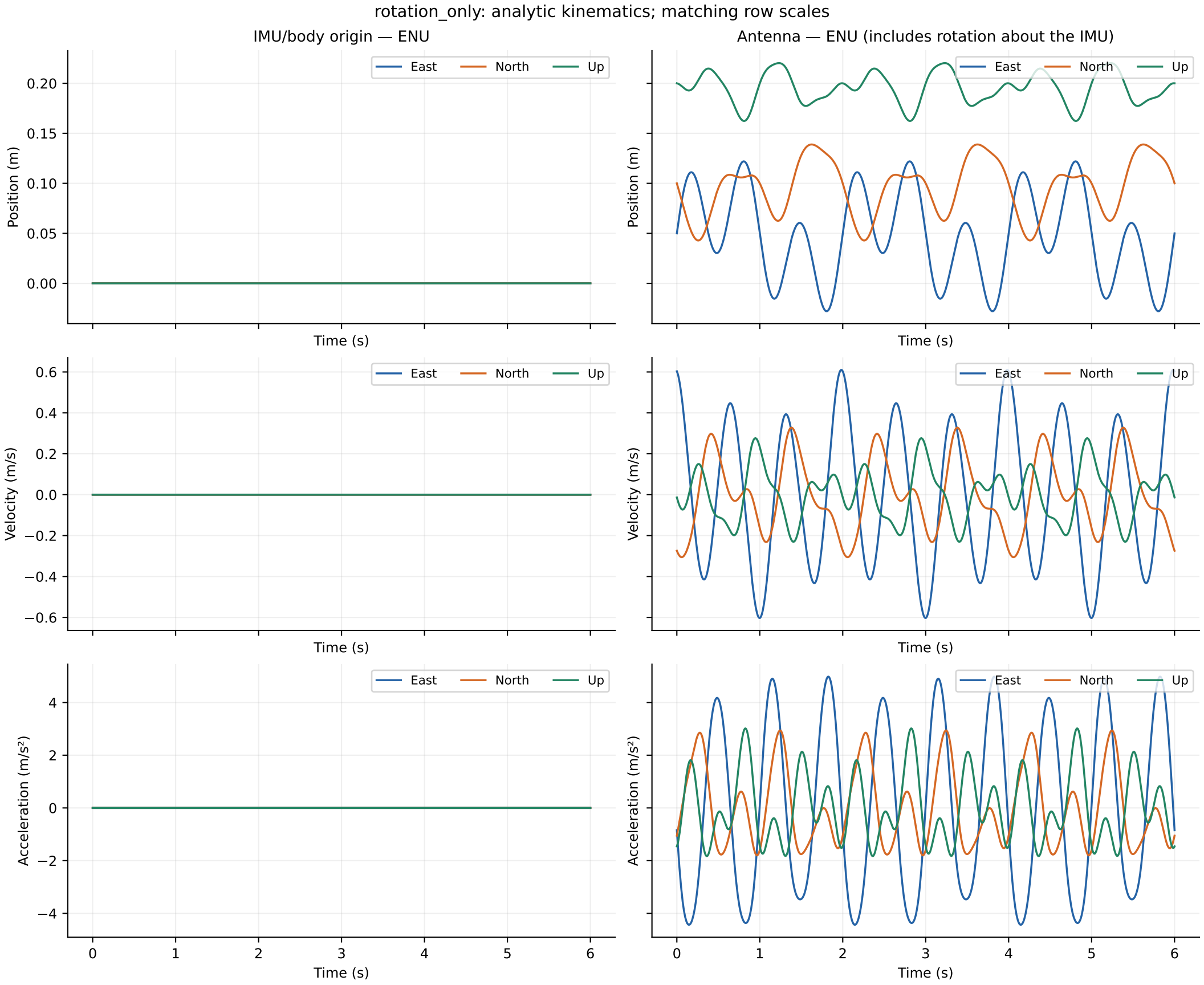


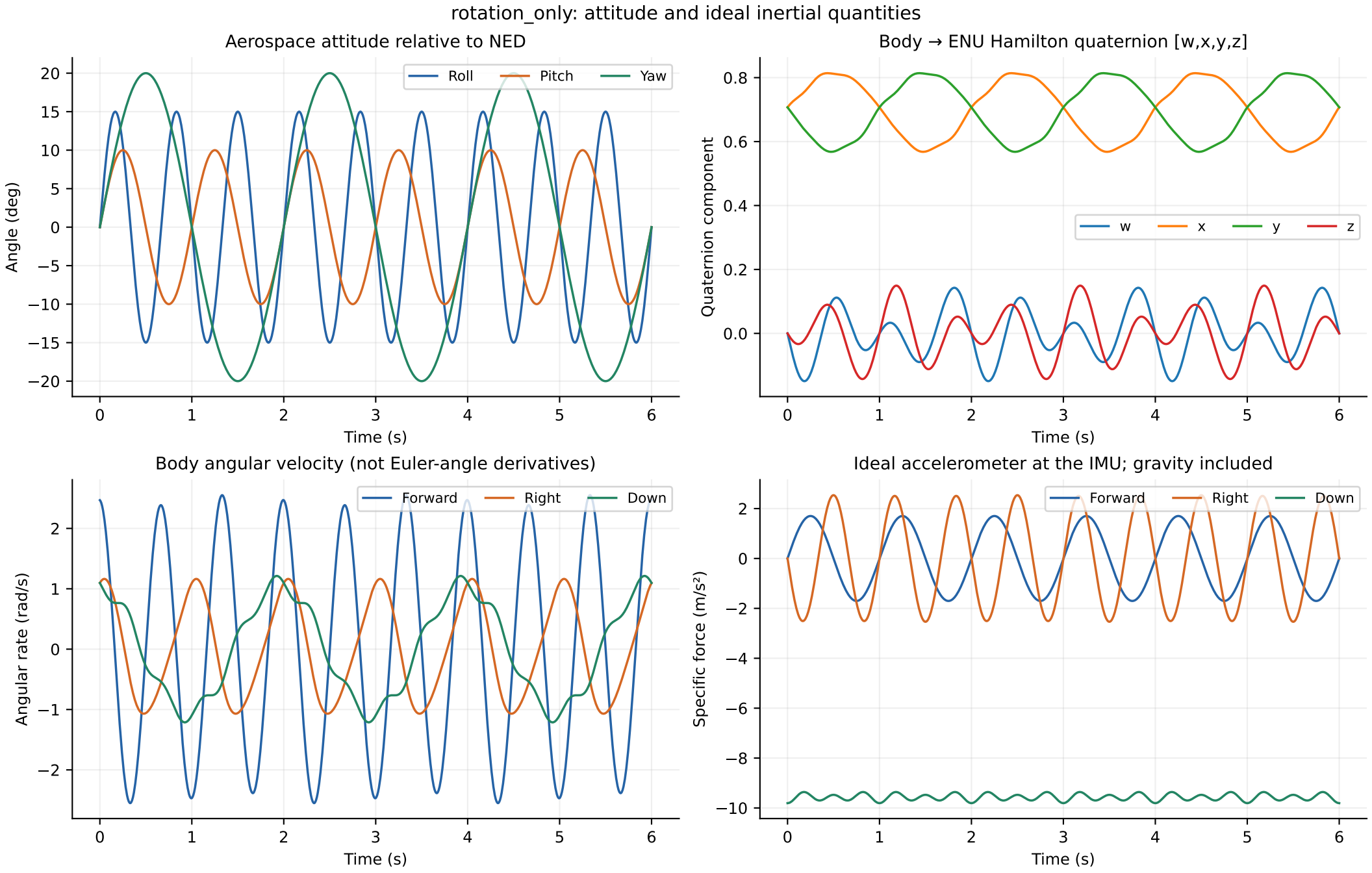


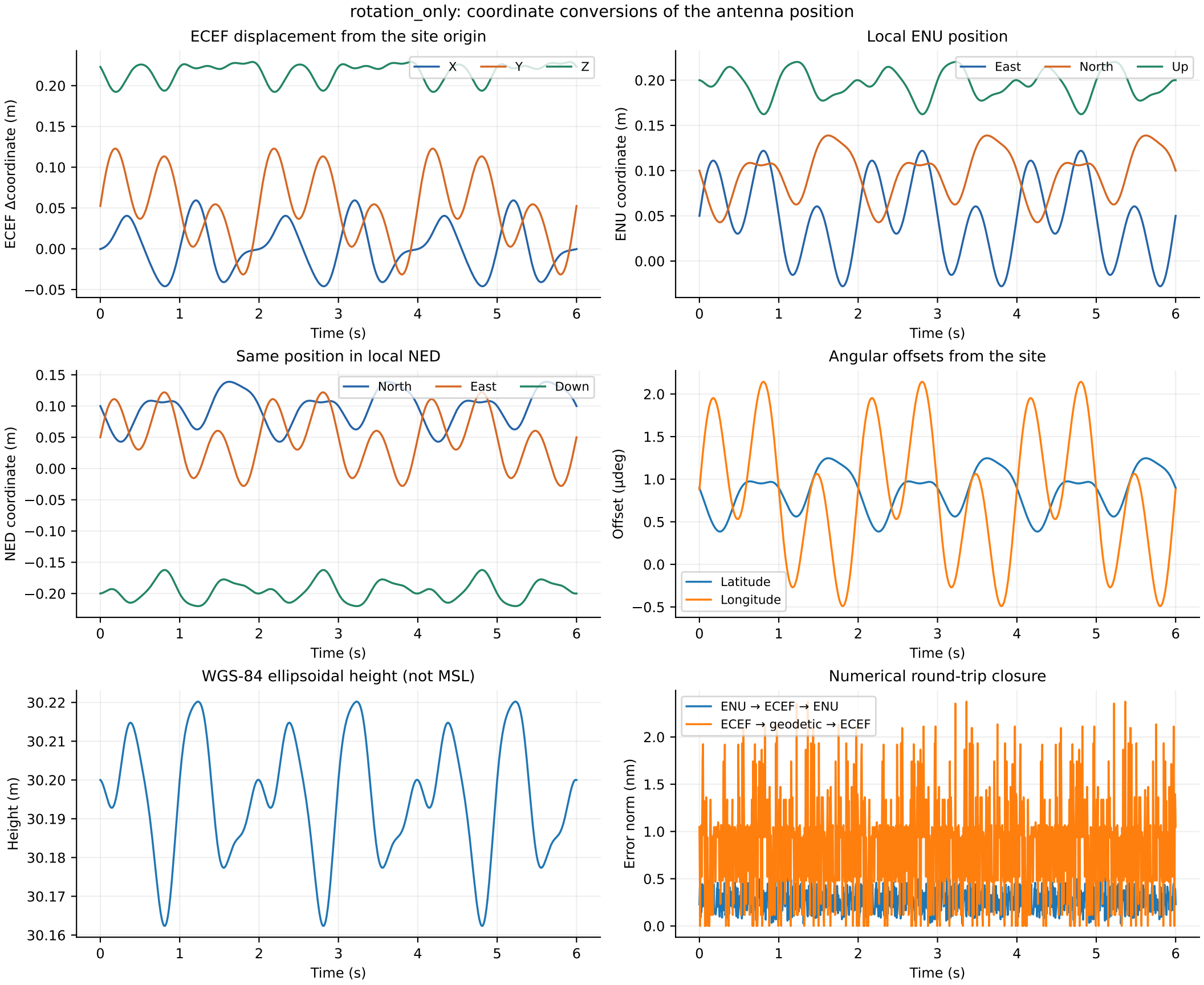


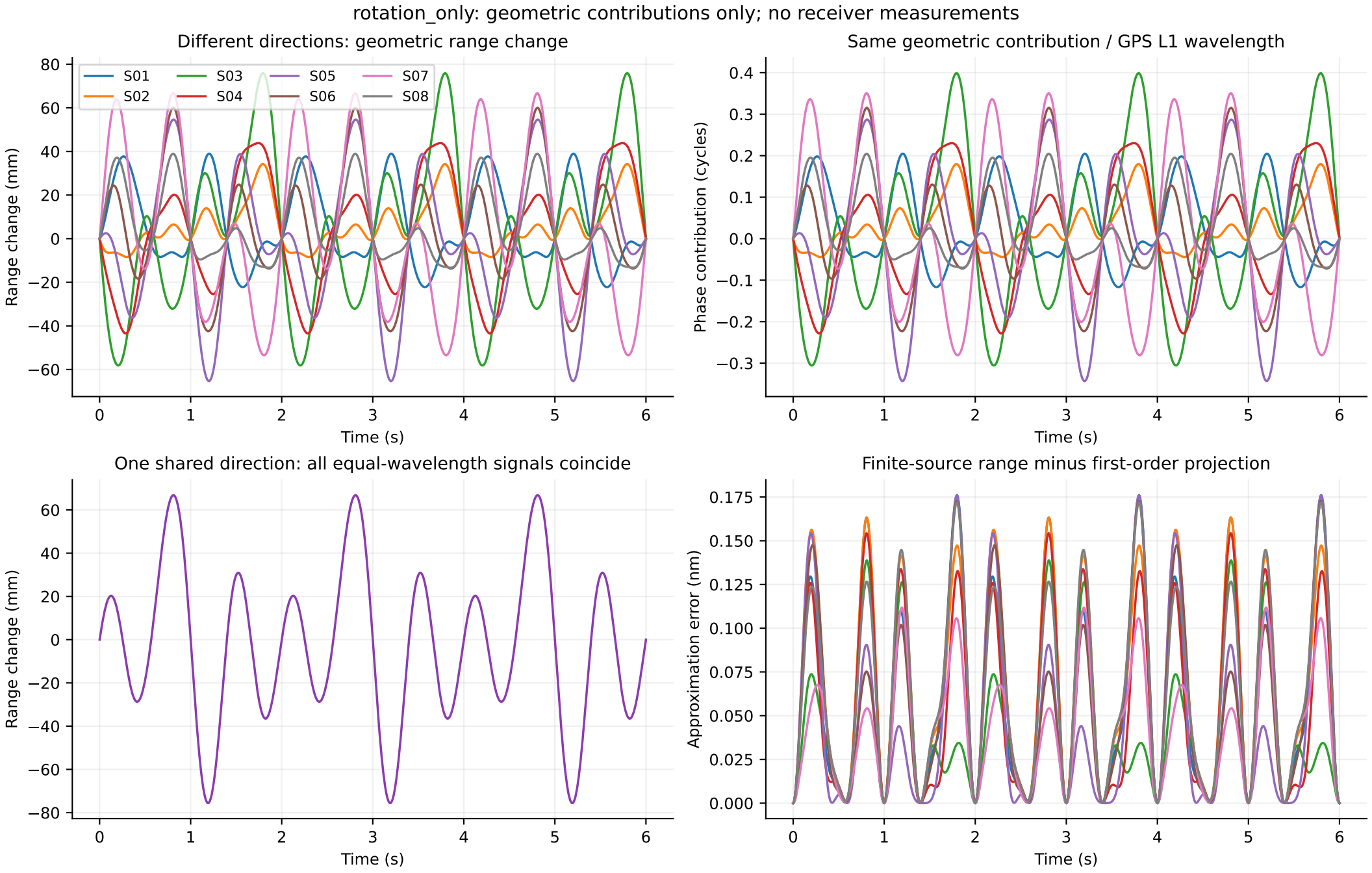

three_axis: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
S02
S03
S04
S05
S06
S07
S08
Same geometric contribution / GPS L1 wavelength
Phase contribution (cycles)
Time (s)
One shared direction: all equal-wavelength signals coincide
Range change (mm)
Time (s)
Finite-source range minus first-order projection
Approximation error (nm)
Time (s)

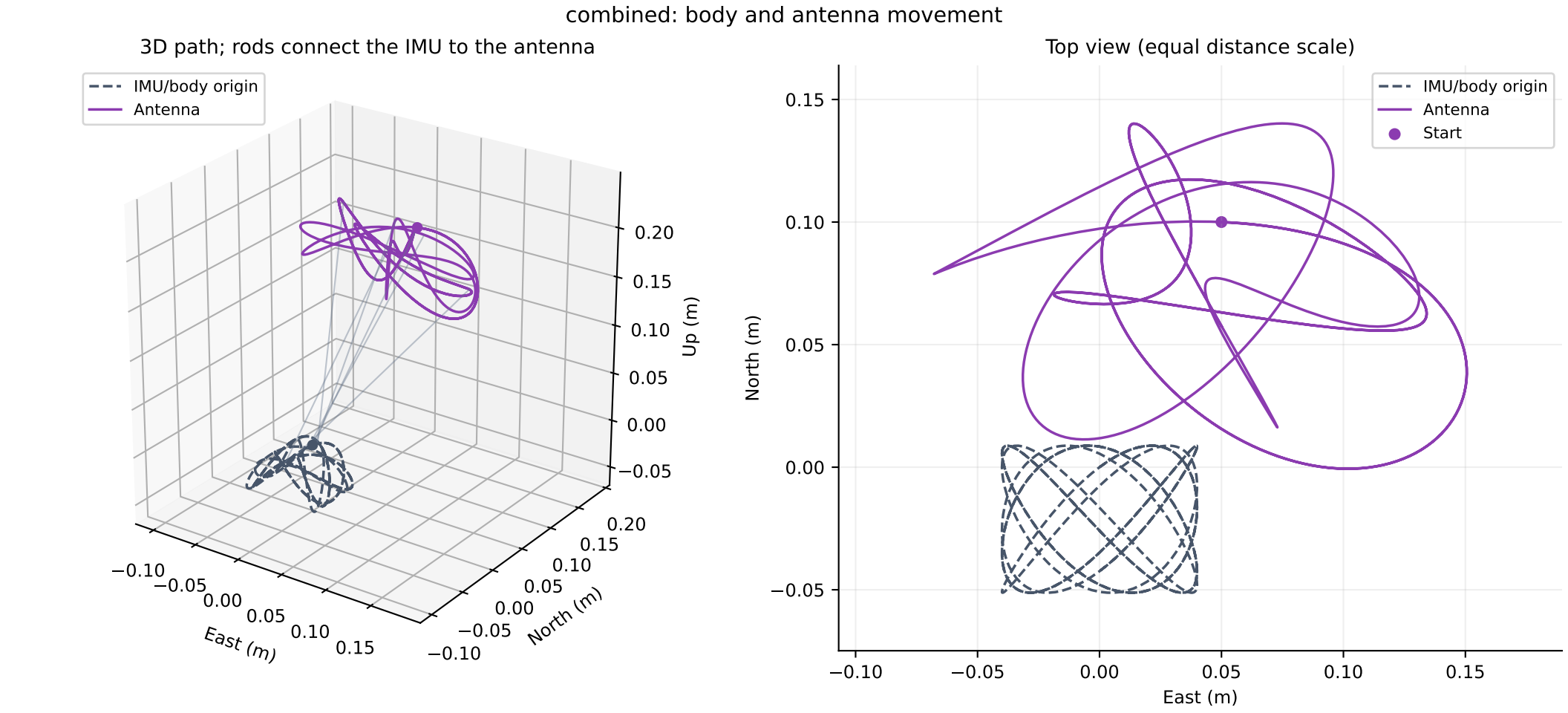

rotation_only: body and antenna movement
3D path; rods connect the IMU to the antenna
IMU/body origin
Antenna
East (m)
North (m)
Up (m)
Top view (equal distance scale)
Start
North (m)
East (m)

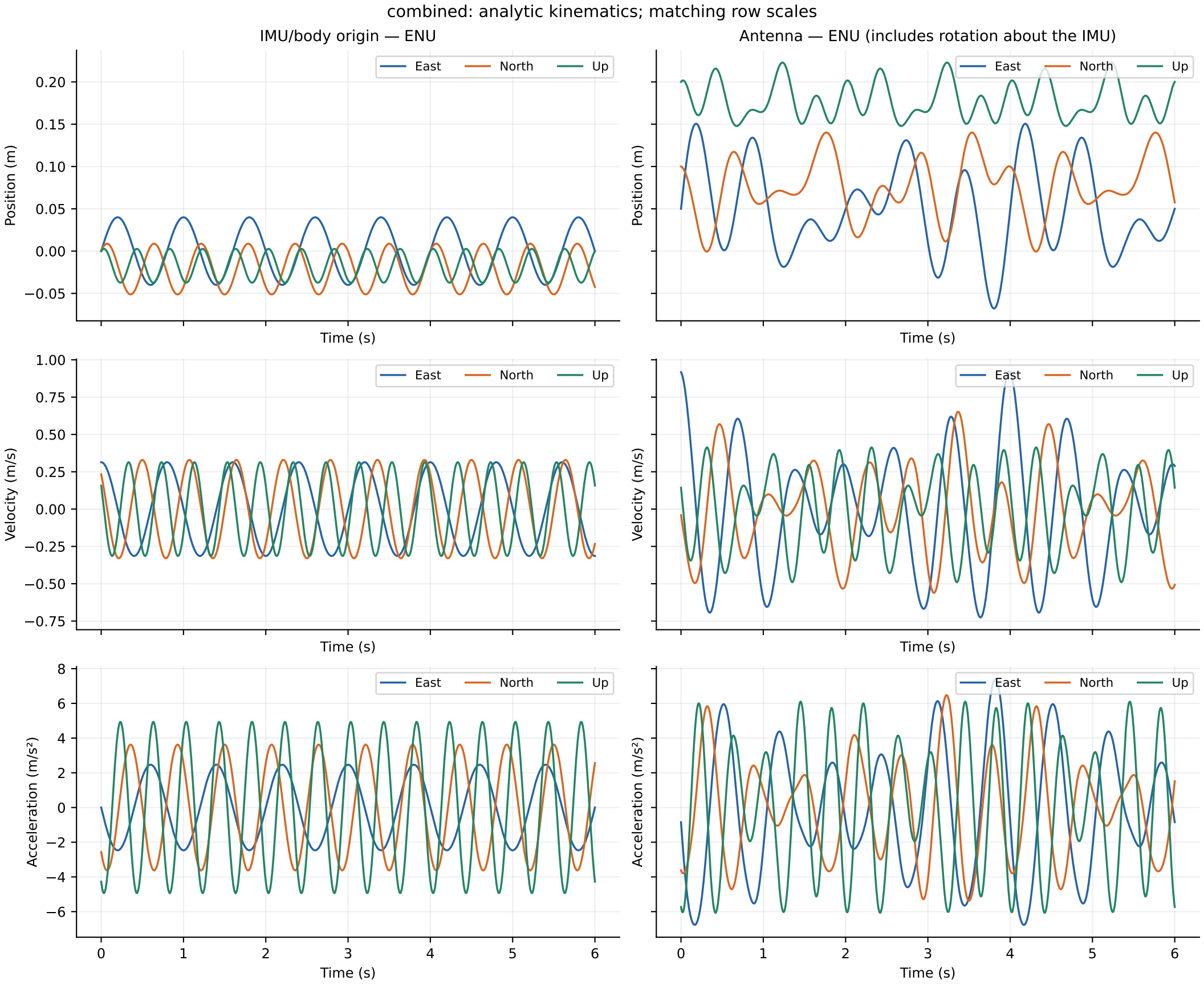

rotation_only: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)
0.20
0.15
0.10
0.05
0.00
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
4
2
0
−2
−4
0
1
2
3
4
5
6

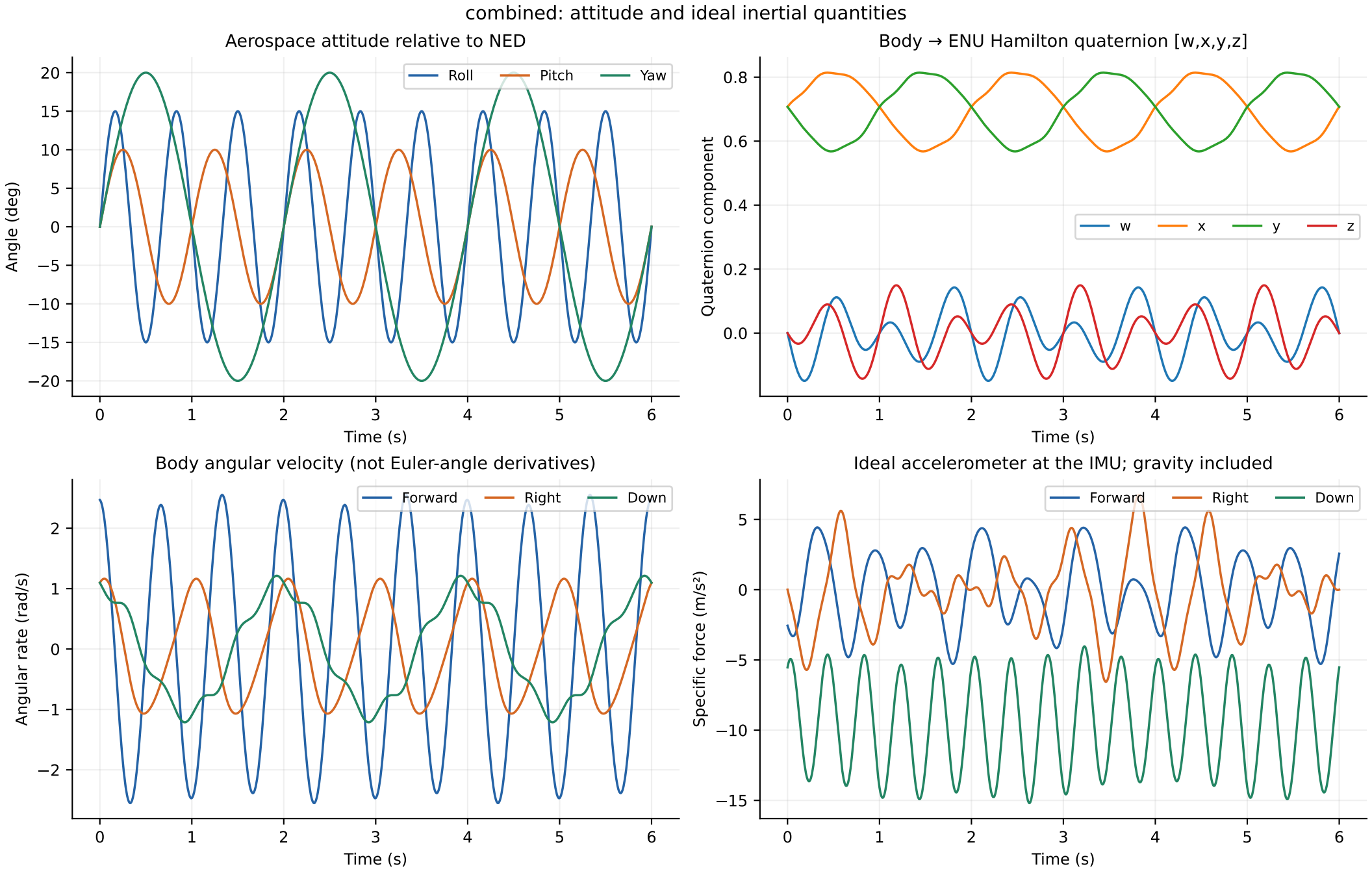

rotation_only: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
Time (s)
Body → ENU Hamilton quaternion [w,x,y,z]
w
x
y
z
Quaternion component
Time (s)
Body angular velocity (not Euler-angle derivatives)
Forward
Right
Down
Angular rate (rad/s)
Time (s)
Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)

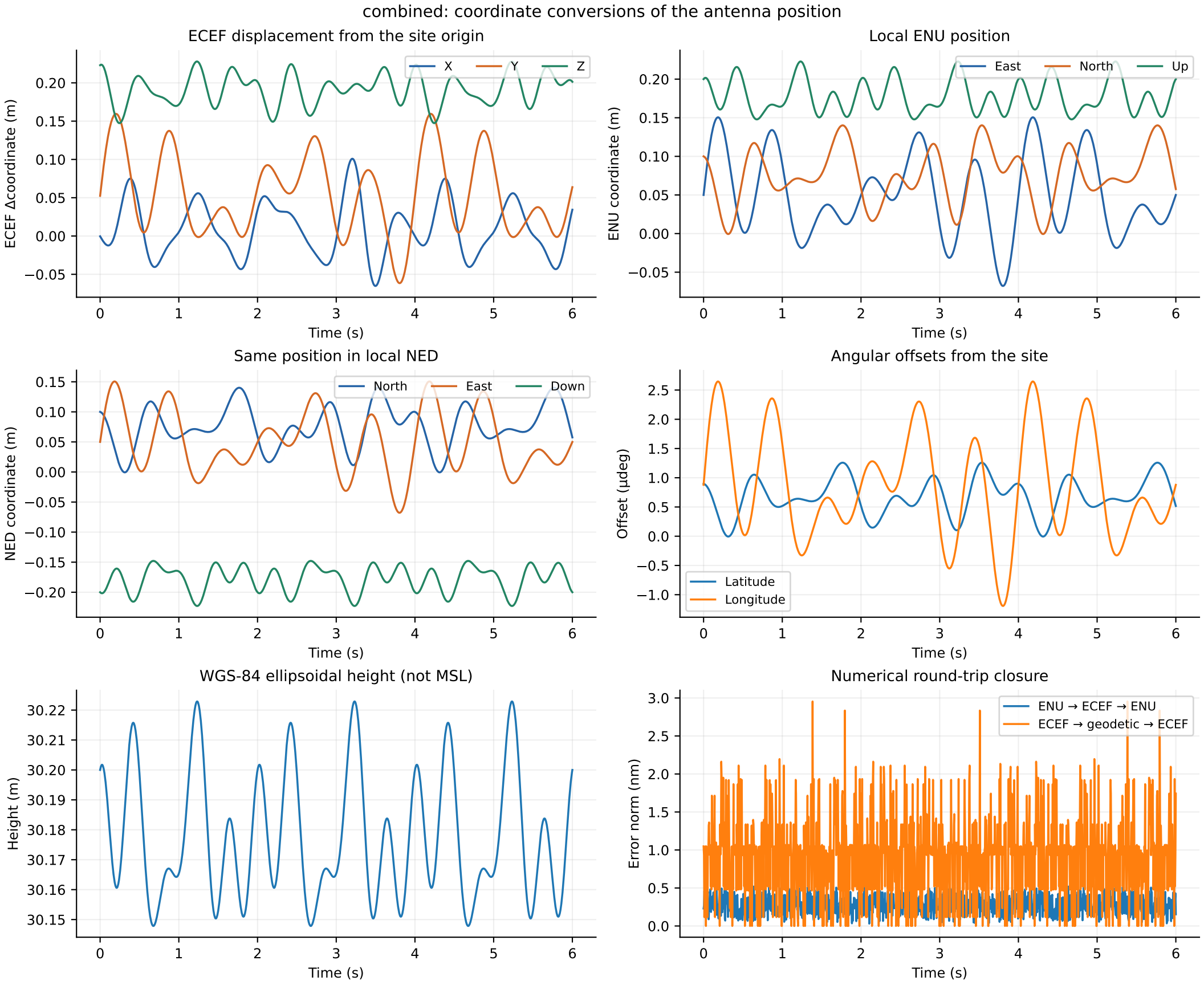

rotation_only: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (µdeg)
Latitude
Longitude
Time (s)
WGS-84 ellipsoidal height (not MSL)
Height (m)
Time (s)
Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)

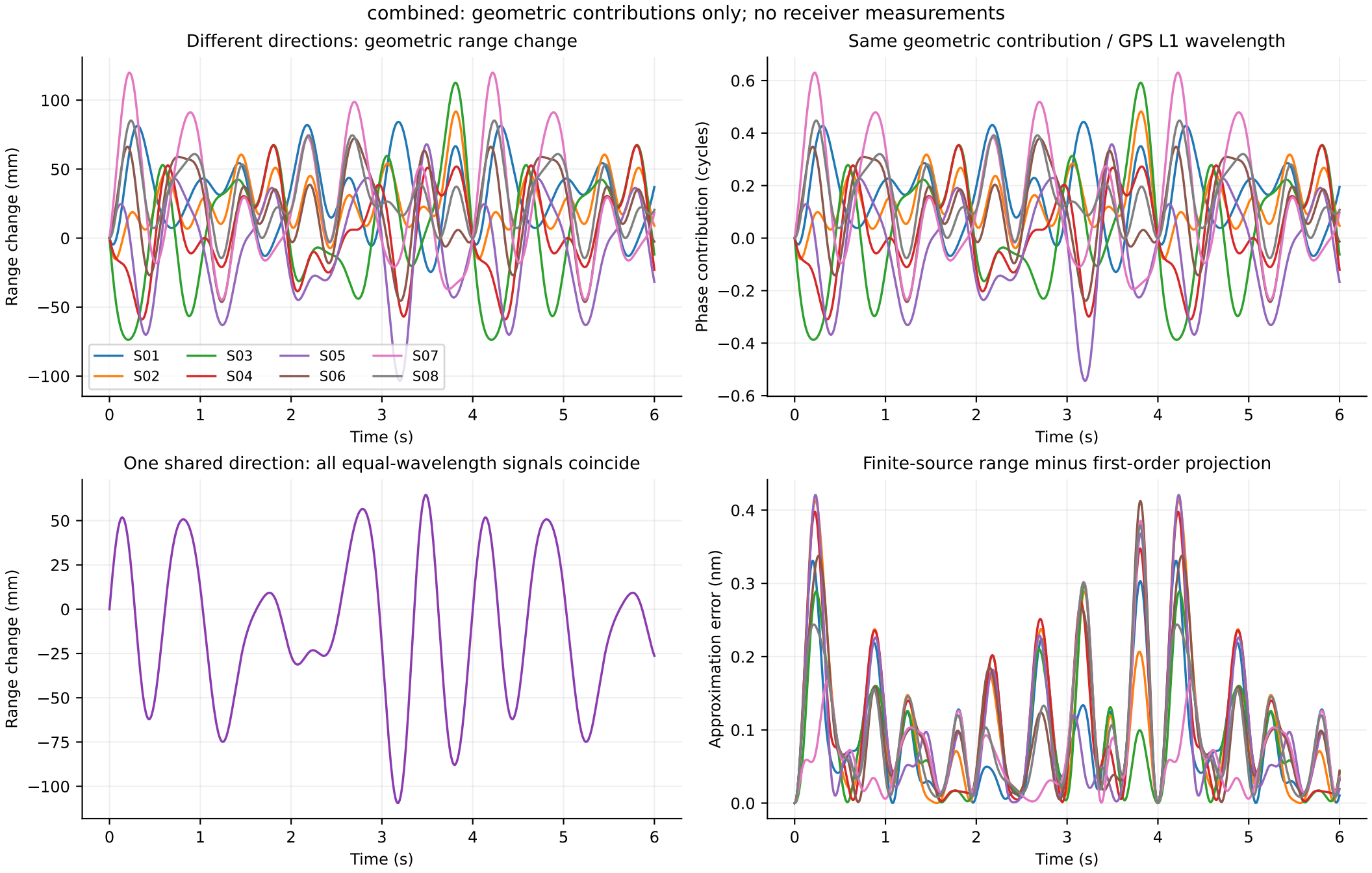

rotation_only: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
S02
S03
S04
S05
S06
S07
S08
Same geometric contribution / GPS L1 wavelength
Phase contribution (cycles)
Time (s)
One shared direction: all equal-wavelength signals coincide
Range change (mm)
Time (s)
Finite-source range minus first-order projection
Approximation error (nm)
Time (s)


combined: body and antenna movement
3D path; rods connect the IMU to the antenna
IMU/body origin
Antenna
East (m)
North (m)
Up (m)
Top view (equal distance scale)
Start
North (m)
East (m)


combined: analytic kinematics; matching row scales
IMU/body origin — ENU
Antenna — ENU (includes rotation about the IMU)
East
North
Up
Position (m)
Velocity (m/s)
Acceleration (m/s²)
Time (s)


combined: attitude and ideal inertial quantities
Aerospace attitude relative to NED
Roll
Pitch
Yaw
Angle (deg)
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Angular rate (rad/s)
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Ideal accelerometer at the IMU; gravity included
Forward
Right
Down
Specific force (m/s²)
Time (s)


combined: coordinate conversions of the antenna position
ECEF displacement from the site origin
ECEF Δcoordinate (m)
X
Y
Z
Time (s)
Local ENU position
ENU coordinate (m)
East
North
Up
Time (s)
Same position in local NED
NED coordinate (m)
North
East
Down
Time (s)
Angular offsets from the site
Offset (µdeg)
Latitude
Longitude
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Height (m)
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Numerical round-trip closure
Error norm (nm)
ENU → ECEF → ENU
ECEF → geodetic → ECEF
Time (s)


combined: geometric contributions only; no receiver measurements
Different directions: geometric range change
Range change (mm)
Time (s)
S01
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S08
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